

In the previous chapter, we illustrated how to flexibly model and estimate the effect of a continuous covariate z on the response variable y without specifying a restrictive functional form of the effect $f(z)$. We also showed how to extend the concepts for two continuous covariates z_1 and z_2 or for a location variable s . As shown in Chap. 3 through various examples, a linear effect was reasonable for a considerable number of the available covariates. However, there are often one or more continuous covariates $z_1, \dots, z_q$, whose effects cannot, at least not a priori, be described with a simple functional form. Hence, we are generally interested in flexibly modeling the effect of these covariates in form of a function $f(z_1, \dots, z_q)$. As seen in Sect. 8.3, estimation of these high-dimensional functions is problematic and requires a very large sample size (to circumvent the curse of dimensionality). Therefore, we often assume a more restrictive additive structure for the effect of the covariates, i.e.,

$$f(z_1, \dots, z_q) = f_1(z_1) + \dots + f_q(z_q).$$

Moreover, models with nonlinear interaction effects or geosadditive models, which consider spatial information in form of a location variable, or models with random effects are also of interest. Sections 9.1–9.4 discuss these models for standard regression problems with continuous responses and approximately normally distributed errors. The models can be formulated in a unified framework as structured additive regression models and can be generalized for non-normal and discrete response variables; compare Sects. 9.5 and 9.6. We will primarily concentrate on the penalization methods and the corresponding Bayesian approaches that were outlined in Sects. 8.1 and 8.2, as they allow for unified estimation of the model parameters, but will also consider boosting in Sect. 9.7. Section 9.8 presents an extensive case study to summarize various aspects of regression modeling in general and structured additive regression in particular.

9.1 Additive Models

We assume a data setting which is similar to that of linear regression. Consider observations $(y_i, x_{i1}, \dots, x_{ik})$, $i = 1, \dots, n$, of a continuous response variable y and covariates $x_1, \dots, x_k$, whose effect on y can be modeled through a linear predictor. Additionally, we have observations $(z_{i1}, \dots, z_{iq})$, $i = 1, \dots, n$, of *continuous* covariates $z_1, \dots, z_q$ whose effects are to be modeled and analyzed nonparametrically. For instance in the rent index application, y represents the net rent per square meter, z_1 the living area, z_2 the year of construction, and the covariates $x_1, \dots, x_k$ the binary coded specific characteristics of the apartment, as well as the experts' assessment of the location in three categories: average, good, and top location.

Additive models are defined by

$$\begin{aligned} y_i &= f_1(z_{i1}) + \dots + f_q(z_{iq}) + \beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik} + \varepsilon_i \\ &= f_1(z_{i1}) + \dots + f_q(z_{iq}) + \eta_i^{lin} + \varepsilon_i \\ &= \eta_i^{add} + \varepsilon_i \end{aligned} \quad (9.1)$$

with

$$\eta_i^{lin} = \beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik}, \quad \eta_i^{add} = f_1(z_{i1}) + \dots + f_q(z_{iq}) + \eta_i^{lin}.$$

The functions $f_1(z_1), \dots, f_q(z_q)$ are nonlinear smooth effects of the covariates $z_1, \dots, z_q$, and are modeled and estimated in a nonparametric way (see Sect. 9.6). In principle, all scatter plot smoothers from Sect. 8.1 can be used as building blocks. At this point, however, we limit ourselves to penalization methods and their corresponding Bayesian approaches.

As already mentioned in Sect. 2.6, additive models have an *identification problem*. More specifically, if we add a constant $a \neq 0$ to the function $f_1(z_1)$ and subtract at the same time a from a second function $f_2(z_2)$, the sum

$$f_1(z_1) + f_2(z_2) = f_1(z_1) + a + f_2(z_2) - a$$

remains the same, i.e., the predictor η does not change if $f_1(z_1)$ changes to $\tilde{f}_1(z_1) = f_1(z_1) + a$ and $f_2(z_2)$ changes to $\tilde{f}_2(z_2) = f_2(z_2) - a$. Although the functional form of $f_1(z_1)$ and $\tilde{f}_1(z_1)$ or $f_2(z_2)$ and $\tilde{f}_2(z_2)$ is unchanged, the overall level of the functions is arbitrary without imposing further restrictions. Hence it is necessary to fix the level of the functions. This is usually obtained by “centering the functions around zero,” such that

$$\sum_{i=1}^n f_1(z_{i1}) = \dots = \sum_{i=1}^n f_q(z_{iq}) = 0$$

holds.

9.1 Standard Additive Regression Model

Data

$(y_i, z_{i1}, \dots, z_{iq}, x_{i1}, \dots, x_{ik}), i = 1, \dots, n$, with y and $x_1, \dots, x_k$ as in the linear regression model and additional continuous covariates $z_1, \dots, z_q$.

Model

$$y_i = f_1(z_{i1}) + \dots + f_q(z_{iq}) + \beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik} + \varepsilon_i.$$

The functions $f_1(z_1), \dots, f_q(z_q)$ are assumed to be smooth nonlinear effects of the continuous covariates $z_1, \dots, z_q$. The same assumptions are made for the errors ε_i as with the classical linear model.

For the standard additive model, the same error assumptions are made as in the classical linear model, i.e., the errors ε_i are i.i.d. and normally distributed with $E(\varepsilon_i) = 0$, $\text{Var}(\varepsilon_i) = \sigma^2$. As with the linear model, the assumptions regarding the error variables ε_i carry over to the response variables y_i , i.e., the y_i are (conditionally) independent with

$$\begin{aligned} E(y_i) &= \mu_i = f_1(z_{i1}) + \dots + f_q(z_{iq}) + \beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik}, \\ \text{Var}(y_i) &= \sigma^2 \end{aligned}$$

and, if applicable, normally distributed with

$$y_i \sim N(\mu_i, \sigma^2).$$

Analogous to linear models, more general assumptions, e.g., heteroscedastic errors, are possible. However, we do not pursue these extensions here.

The literature often refers the special case

$$y_i = f_1(z_{i1}) + \dots + f_q(z_{iq}) + \varepsilon_i, \quad (9.2)$$

without the additional linear predictor η_i^{lin} as the additive model. The model (9.1) is then called *partial linear model* or *semiparametric model*.

In either case, additive models (9.1) or (9.2) do not contain interactions between covariates. Models with interactions will be discussed in Sect. 9.3.

The nonlinear and linear effects can be interpreted as described already in Sect. 2.6. Further illustration is provided with the rent index data, as we will see in Example 9.1 below.

For modeling and estimation of nonlinear functions, this chapter primarily focuses on penalization and Bayesian methods with basis functions as summarized in Sect. 8.1.3. Hence, the functions f_j , $j = 1, \dots, q$, will be approximated by

$$f_j(z_j) = \sum_{l=1}^{d_j} \gamma_{jl} B_l(z_j),$$

where the basis functions B_l represent TP- or B-spline bases of polynomial splines, smoothing splines, or even kriging methods. In principle, it is possible to choose different types of basis functions for the different functions f_j , e.g., polynomial splines of different order. For simplicity in presentation, such variations are notationally suppressed. The vector $\mathbf{f}_j = (f_j(z_{1j}), \dots, f_j(z_{nj}))'$ of function values evaluated at the observed covariate values $z_{1j}, \dots, z_{nj}$ can then be expressed as

$$\mathbf{f}_j = \mathbf{Z}_j \boldsymbol{\gamma}_j,$$

where $\boldsymbol{\gamma}_j = (\gamma_{j1}, \dots, \gamma_{jd_j})'$ is the vector of regression coefficients. The design matrix $\mathbf{Z}_j$ consists of the basis functions evaluated at the observed covariate values, i.e., $\mathbf{Z}_j[i, l] = B_l(z_{ij})$.

As in the linear model, $\mathbf{y} = (y_1, \dots, y_n)'$ defines the vector of the response values, $\boldsymbol{\varepsilon} = (\varepsilon_1, \dots, \varepsilon_n)'$ the vector of the error values, $\boldsymbol{\beta} = (\beta_0, \dots, \beta_k)'$ the vector of regression coefficients for the linear part of the predictor, and $\mathbf{X}$ the corresponding design matrix. The additive model can then be written in matrix notation in the form

$$\mathbf{y} = \mathbf{Z}_1 \boldsymbol{\gamma}_1 + \dots + \mathbf{Z}_q \boldsymbol{\gamma}_q + \mathbf{X} \boldsymbol{\beta} + \boldsymbol{\varepsilon}. \quad (9.3)$$

At first glance, Eq. (9.3) appears to be a large linear model. In fact, this would be the case if the coefficient vectors $\boldsymbol{\gamma}_1, \dots, \boldsymbol{\gamma}_q$ of basis functions are estimated, unrestrictedly, using (ordinary) least squares. In general, however, estimation is regularized by penalties of the form $\lambda_j \boldsymbol{\gamma}_j' \mathbf{K}_j \boldsymbol{\gamma}_j$ (or corresponding smoothness priors), as described for a single function f in Sect. 8.1.3, to enforce specific smoothness properties of the estimates and to achieve regularization. The form of the penalty matrix depends on the specific choice of basis functions. Details on how to estimate the model will be provided in Sect. 9.6.

As in the example to follow, we will use penalized splines (Sect. 8.1.2) or the corresponding Bayesian version, in which the Gaussian prior $p(\boldsymbol{\gamma}_j | \tau_j^2) \propto \exp(-0.5 \boldsymbol{\gamma}_j' \mathbf{K}_j \boldsymbol{\gamma}_j / \tau_j^2)$ replaces the penalty function.

Example 9.1 Munich Rent Index—Additive Model



We choose the net rent per square meter as the response variable. As covariates we include the living area (*area*), the year of construction (*yearc*), and the location (*location*). This leads to the additive model

$$\text{rentsqm} = f_1(\text{area}) + f_2(\text{yearc}) + \beta_0 + \beta_1 g_{\text{location}} + \beta_2 t_{\text{location}} + \varepsilon.$$

9.2 Basis Function Approach to Additive Regression

The functions f_j are approximated using basis functions of the form

$$f_j(z_j) = \sum_{l=1}^{d_j} \gamma_{jl} B_l(z_j).$$

Regardless of the chosen basis, we obtain the additive model

$$y = \mathbf{Z}_1 \boldsymbol{\gamma}_1 + \dots + \mathbf{Z}_q \boldsymbol{\gamma}_q + \mathbf{X} \boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

where the design matrices $\mathbf{Z}_1, \dots, \mathbf{Z}_q$ consist of the basis functions evaluated at the given covariate values. The design matrix $\mathbf{X}$ is constructed as in the linear model.

The functions f_1 and f_2 are modeled by P-splines with 20 interior knots and second-order difference penalties. For the location, the model includes the dummy variables *glocation* for good locations and *tlocation* for top location. The average location serves as the reference category. Note that the model is a pure main effects model, i.e., we do not consider possible interactions, e.g., between living area and location. As mentioned, such interaction models are discussed in Sect. 9.3.

Figure 9.1 shows the estimated nonlinear effects of living area and year of construction. The results are obtained using `remlreg` objects of the software `BayesX`. In the left panels the partial residuals are additionally included. Their calculation is completely analogous to partial residuals in linear models described in section “Statistical Properties of Residuals” of Sect. 3.2.3. For example, the partial residuals with regard to the apartment’s living area are defined by

$$\hat{\varepsilon}_{area,i} = \hat{\varepsilon}_i + \hat{f}_1(area_i).$$

Also see Sect. 9.5 for further details regarding partial residuals in additive models.

As expected, the estimated function of the living area shows a nonlinear decreasing effect on rent with increasing area. For living area greater than about 110 sqm, the width of pointwise confidence intervals increases due to fewer larger apartments. Consequently, we should not over-interpret the slightly increasing effect at the right margin: A horizontal trend, from about 110 sqm onward, easily fits within the confidence limits. Initially, the estimated effect for the year of construction is almost constant. However, from 1945 onward, the effect is almost linearly increasing. All in all, the estimated nonlinear effects are very similar to those found in the parametric model of Example 3.5 (p. 90) in Chap. 3. Nevertheless, the fit to the data is better with the additive model used here. If we rather use the transformation $1/area$ for the living area and a cubic polynomial for modeling the effect of the year of construction, as presented in Example 3.5, we obtain an AIC of 7,380.3. With the nonparametric model, we obtain a lower AIC of 7,362.6 such that the nonparametric model is preferable. The calculation of the AIC and other goodness-of-fit measures for additive models are detailed in Sect. 9.5. In contrast to the parametric model of Example 3.5, the nonparametric approach has yet another distinct advantage: Nonlinear effects are obtained in an automatic way, i.e., it is not necessary to reflect upon, e.g., the type of variable transformation or the order of the polynomial.

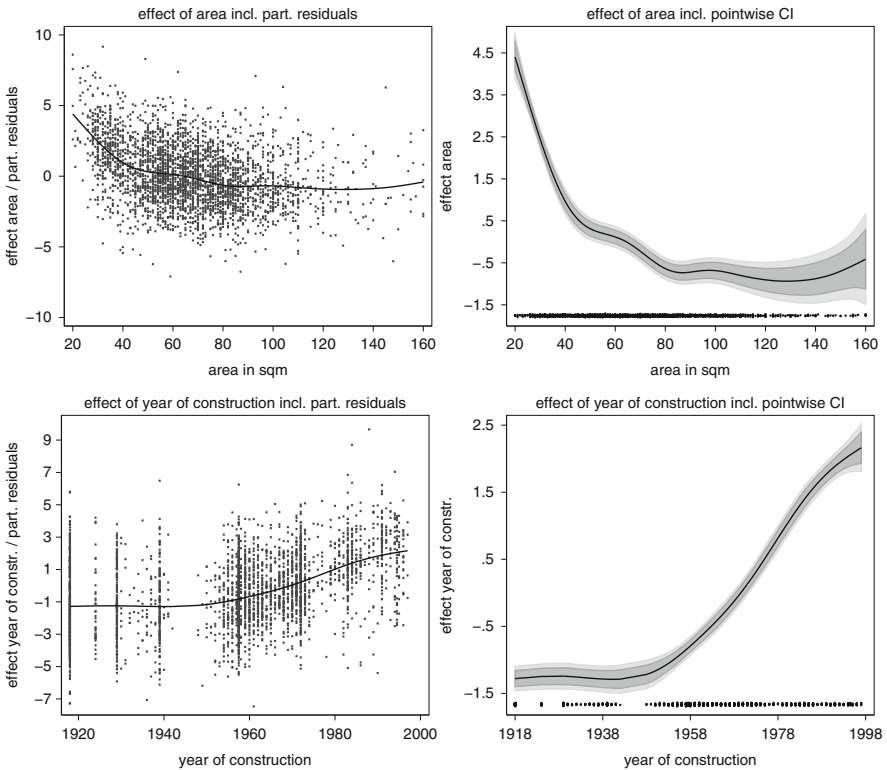


Fig. 9.1 Munich rent index: estimated nonlinear effects of area and year of construction

Table 9.1 Munich rent index: estimated effects of location

Variable	Coefficient	Standard error	p-value	95 % confidence interval	
<i>intercept</i>	7.083	0.120	<0.001	6.848	7.317
<i>glocation</i>	0.633	0.076	<0.001	0.484	0.783
<i>llocation</i>	1.503	0.233	<0.001	1.046	1.961

Finally, we observe comparably strong and highly significant additional charge of approximately 0.63 Euro for a good location and of approximately 1.5 Euro for top location, compared to average location (see Table 9.1). △

9.2 Geoadditive Regression

In addition to the three-categorical expert assessment of location, the rent index data also contain information in which of the 336 districts in Munich the apartment is located. If we want to use the more precise location variable *district* rather than the expert assessment, we arrive at the geoadditive model

$$\text{rentsqm} = \beta_0 + f_1(\text{area}) + f_2(\text{yearc}) + f_{\text{geo}}(\text{district}) + \varepsilon.$$

It is also possible to additionally include the expert assessment in the model. In this case, we obtain the modified geoadditive model

$$\begin{aligned} \text{rentsqm} = & f_1(\text{area}) + f_2(\text{yearc}) + f_{\text{geo}}(\text{district}) + \beta_0 + \beta_1 \text{glocation} \\ & + \beta_2 \text{tlocation} + \varepsilon. \end{aligned}$$

In general, the data for geoadditive models are given through observations $(y_i, \mathbf{x}_i, \mathbf{z}_i)$, $i = 1, \dots, n$, for the response variable y and the covariates $\mathbf{x}, \mathbf{z}$ (as in the additive model), plus the additional values s_i , $i = 1, \dots, n$, associated with a geographic location index s . As already outlined in Sect. 8.2, the variable s is either discrete, taking the values $s \in \{1, \dots, d\}$ (see Sect. 8.2.4), or varies continuously in a subset of $\mathbb{R}^2$ (see Sects. 8.2.1–8.2.3). In the discrete case, s_i defines a specific region, in which the individual or the unit i was observed, e.g., the district in Munich, districts in Zambia, counties, and area code districts in Germany. In the continuous case, s is usually given by the coordinates of $\mathbb{R}^2$, i.e., the location s_i of the unit i is (almost) exactly known. In the original sample of the Munich rent index, the exact address of the apartment is provided. However, due to data confidentiality, we are only given the district in which the apartment is located. Data derived from insurance policies are handled in a similar way. The insurance companies know the exact address of the apartment, but due to data confidentiality, this data is aggregated to a discrete location variable, for example, for counties or licensing districts.

In geoadditive models, the predictor is extended by the spatial effect $f_{\text{geo}}(s)$ of the location variable s , and we obtain

$$\begin{aligned} y_i &= \eta_i^{\text{add}} + f_{\text{geo}}(s_i) + \varepsilon_i \\ &= f_1(z_{i1}) + \dots + f_q(z_{iq}) + f_{\text{geo}}(s_i) + \mathbf{x}_i' \boldsymbol{\beta} + \varepsilon_i, \end{aligned} \quad (9.4)$$

with $\mathbf{x}_i' \boldsymbol{\beta} = \beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik}$. For the covariates $\mathbf{x}_i, \mathbf{z}_i$ and the error variables ε_i , the same assumptions apply as for the additive model.

The spatial effect $f_{\text{geo}}(s)$ can be understood as a surrogate for unobserved spatial variables not included in the data. The approaches presented in Sect. 8.2 can be used as building blocks for the modeling and estimation of $f_{\text{geo}}(s)$. Markov random fields (see Sect. 8.2.4) are especially useful for discrete $s \in \{1, \dots, d\}$. It is also possible to apply smoothers for continuous s , in particular kriging (see Sect. 8.2.3), e.g., by using the region centroids as coordinates.

For discrete locations, $s_i = s$ denotes that the i th observation belongs to region s , with $s \in \{1, \dots, d\}$, then $f_{\text{geo}}(s_i)$ is the spatial effect of region $s = s_i$. Similar to the basis function approaches discussed in the previous section, we can represent the vector $\mathbf{f}_{\text{geo}} = (f_{\text{geo}}(s_1), \dots, f_{\text{geo}}(s_n))'$ of the spatial effect for the observed units $i = 1, \dots, n$, as

$$\mathbf{f}_{\text{geo}} = \mathbf{Z}_{\text{geo}} \boldsymbol{\gamma}_{\text{geo}}. \quad (9.5)$$

Here, $\boldsymbol{\gamma}_{geo} = (\gamma_{geo,1}, \dots, \gamma_{geo,d})'$ is the vector of regression coefficients and the $n \times d$ design matrix $\mathbf{Z}_{geo}$ is an incidence matrix, i.e., with $\mathbf{Z}_{geo}[i, s] = 1$ for the s th element of the i th row if $s_i = s$, and 0 otherwise. For a continuous s , $\boldsymbol{f}_{geo}$ can also be written in the form (9.5). The specific forms of $\mathbf{Z}_{geo}$ and $\boldsymbol{\gamma}_{geo}$ can be found in Sects. 8.2.1–8.2.3. When using kriging, we usually work with a representative number of knots (see p. 518).

We can now write geoaddivitive models in matrix notation as

$$\mathbf{y} = \mathbf{Z}_1 \boldsymbol{\gamma}_1 + \dots + \mathbf{Z}_q \boldsymbol{\gamma}_q + \mathbf{Z}_{geo} \boldsymbol{\gamma}_{geo} + \mathbf{X} \boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad (9.6)$$

extending the additive model (9.3) by the spatial effect $\mathbf{Z}_{geo} \boldsymbol{\gamma}_{geo}$.

Similar as for the coefficient vectors $\boldsymbol{\gamma}_1, \dots, \boldsymbol{\gamma}_q$, estimation of $\boldsymbol{\gamma}_{geo}$ is again regularized using quadratic penalties of the form $\lambda_{geo} \boldsymbol{\gamma}_{geo}' \mathbf{K}_{geo} \boldsymbol{\gamma}_{geo}$ or with the Gaussian smoothness priors as described in Sect. 8.2. See Sect. 9.6 for further details.

Example 9.2 Munich Rent Index—Geoaddivitive Model

We consider the two geoaddivitive models

$$rentsqm = f_1(area) + f_2(yearc) + f_{geo}(district) + \beta_0 + \varepsilon \quad (9.7)$$

and

$$rentsqm = f_1(area) + f_2(yearc) + f_{geo}(district) + \beta_0 + \beta_1 glocation + \beta_2 tlocation + \varepsilon \quad (9.8)$$

with the spatial effect $f_{geo}(district)$ of the districts in Munich.

Whereas model (9.7) does not consider the dummies for the expert assessment level of location, they are included within the predictor of model (9.8). In both cases, we choose to use kriging with the district centroids as coordinates for modeling the spatial effect $f_{geo}(district)$. Comparing the estimates $\hat{f}_{geo}$ of the two models allows us to evaluate the expert assessment of location. If the experts assessment regarding the apartment's location is valid the location dummies $glocation$ and $tlocation$ should absorb most of the spatial heterogeneity and the estimated spatial effect $\hat{f}_{geo}$ should be much weaker than in model (9.7).

Figure 9.2 displays the estimated spatial effects. The results are again obtained using `remlreg` objects of `BayesX`. The left panel very well reflects the spatial variation of the rent market in the city of Munich, adjusted to the effects of living area and year of construction. Districts in the southern part of Munich, e.g., Grünwald and Meterschwaige, or those that are close to the Nymphenburg Park and the English Garden, show a rent index that is much higher than the average. Northern districts close to or north of the Frankfurter Ring show a rent index that is much lower than the average. Overall the effect varies between -1 and 1 . An effect of, e.g., 0.5 implies that apartments in this particular subquarter are on average 50 cents more expensive than apartments in subquarters with zero effect (keeping all other covariates constant).

If the location dummies, according to the expert assessment, are additionally included in the predictor, the spatially smooth district effect clearly diminishes. The kernel density estimators of the smooth district effects, illustrated in Fig. 9.3, provide a good impression of the decreased variability. Considering the expert assessment, the kernel density estimator of the district effect shows a higher concentration around zero, but a visible variability still remains. Although we find that the expert assessment absorbs a considerable portion of the spatial heterogeneity, there appears to be opportunity for improvement in expert assessment.

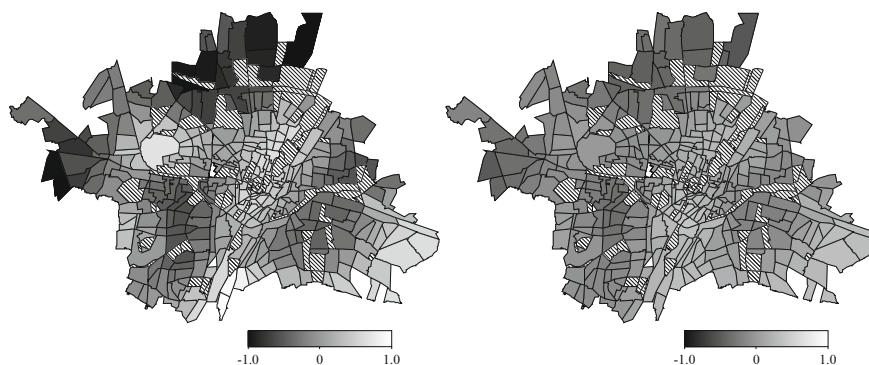


Fig. 9.2 Munich rent index: spatial effect based on kriging. *Left*: expert assessment of location omitted. *Right*: expert assessment of location included

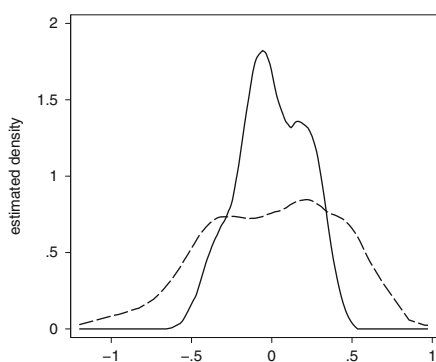


Fig. 9.3 Munich rent index: kernel density estimators for the spatial effect. *Solid line*: model with expert assessment of location included. *Dashed line*: model with expert assessment of location excluded

The quality of the expert assessment of the location can also be seen from the goodness-of-fit measured by the AIC. For model (9.7) we obtain $AIC = 7,359$, while the second model (9.8), yields $AIC = 7,336$. Compared to the model in Example 9.1, where the spatial effect was modeled parametrically with two location dummy variables, the value of the AIC in model (9.7) is only smaller by approximately three units. Only by combining the experts assessment and the nonparametric spatial effect the AIC value significantly decreases by almost 30 units.

△

9.3 Models with Interactions

Section 3.1.3 (p. 98) already illustrated how (pairwise) covariate interactions, along with main effects, can be considered in a linear model. The additive and geoaddivitive models of the previous sections are purely main effect models with respect to the

nonparametric components $f_1, \dots, f_q$ and the spatial effect f_{geo} . In this section, we will show how these models can be further extended using (pairwise) nonparametric interaction terms. At first, we will consider the case of a continuous covariate z and a binary or categorical covariate x . Due to the specific type of interaction, we also speak of models with varying coefficients. Models of this type can also be modified in a way that an interaction between a location variable s and x is possible. Finally, we will address pairwise interactions between two continuous covariates z_1 and z_2 . We use the rent index data for illustration.

9.3.1 Models with Varying Coefficient Terms

In additive models, the interaction between a continuous variable z_1 and a binary variable x_1 can be considered by including the interaction term $f_{z_1|x_1}(z_1) \cdot x_1$. Here, $f_{z_1|x_1}(z_1)$ is a nonparametric smooth function, which similar to the main effect $f_1(z_1)$ needs to be estimated from the data. This results in the model

$$y_1 = f_1(z_{i1}) + \dots + f_q(z_{iq}) + f_{z_1|x_1}(z_{i1}) x_{i1} + \mathbf{x}'_i \boldsymbol{\beta} + \varepsilon_i, \quad (9.9)$$

where the same assumptions apply as for the additive model (9.1). Since the model (9.9) also contains the main effects $f_1(z_1)$ and $\beta_1 x_1$, the function $f_{z_1|x_1}$ must be centered around zero in the same way as the other functions $f_1, \dots, f_q$. Otherwise, the transformations $\tilde{f}_{z_1|x_1} = f_{z_1|x_1}(z_1) + a$ and $\tilde{\beta}_1 = \beta_1 - a$ would result in the same predictor, i.e., the model would not be identifiable. If the main effect $\beta_1 x_1$ is excluded from the model, then centering of $f_{z_1|x_1}$ around zero is not necessary. The varying coefficients model can be interpreted as follows:

- $f_1(z_1)$ is the nonlinear effect of z_1 if $x_1 = 0$.
- $f_1(z_1) + f_{z_1|x_1}(z_1) + \beta_1$ is the nonlinear effect of z_1 if $x_1 = 1$.
- $f_{z_1|x_1}(z_1) + \beta_1$ is a *varying effect* for $x_1 = 1$ (relative to $x_1 = 0$) depending on z_1 . The variable z_1 is also called an *effect modifier* of x_1 , and x_1 is called the *interaction variable*.

Instead of the conventional fixed coefficient β_1 for x_1 , we rather obtain a varying coefficient $f_{z_1|x_1}(z_1) + \beta_1$ depending on z_1 . To get an intuitive understanding about the effects, it is useful to visualize both effects $f_1(z_1)$ and $f_1(z_1) + f_{z_1|x_1}(z_1) + \beta_1$ in a combined graph. Additionally, the varying effect $f_{z_1|x_1}(z_1) + \beta_1$ of x_1 , depending on z_1 , can be plotted in a separate graph.

Similarly, we can define an interaction between z_1 and a three-level categorical variable $x \in \{1, 2, 3\}$, where x is coded by the dummy variables x_1 and x_2 , with the last value 3 as the reference category. This leads to the model

$$y_i = f_1(z_{i1}) + \dots + f_{z_1|x_1}(z_{i1}) x_{i1} + f_{z_1|x_2}(z_{i1}) x_{i2} + \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \varepsilon_i.$$

Interpretation and visualization extends similarly: $f_1(z_1) + f_{z_1|x_1}(z_1) + \beta_1$ is the effect of z_1 if $x = 1$, and $f_1(z_1) + f_{z_1|x_2}(z_1) + \beta_2$ is the effect of z_1 if $x = 2$, whereas $f_1(z)$ is the nonlinear effect of z_1 if $x = 3$. For interpretation, it is again advisable to visualize the effects.

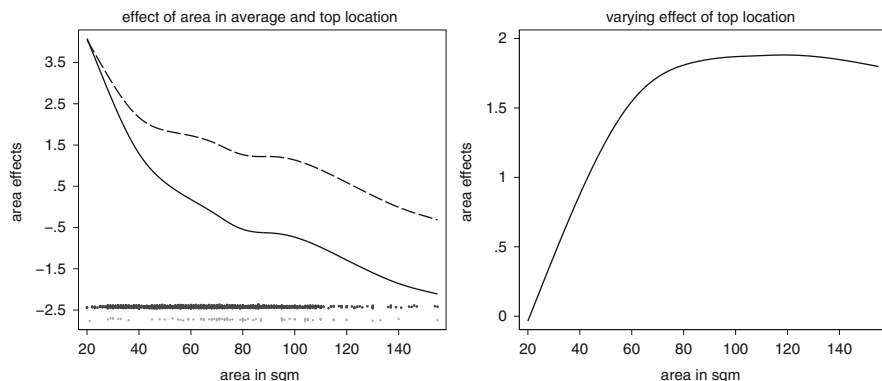


Fig. 9.4 Munich rent index: *left*, effect of area in average (solid line) and top location (dashed line). The dots show the distribution of the observations in average (dark dots) and top location (bright dots). *Right*, varying effect of top location

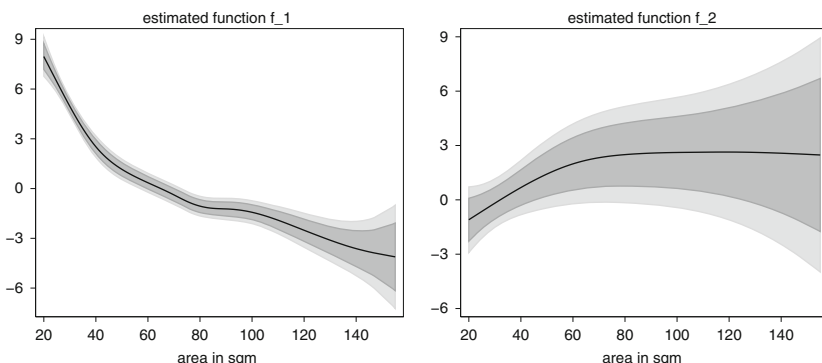


Fig. 9.5 Munich rent index: estimated functions f_1 and f_2 including pointwise confidence intervals in the model (9.10)

Example 9.3 Munich Rent Index—Interaction Between Living Area and Location

This example can be viewed as the nonparametric extension to the strictly parametric Example 3.8 (p. 102). As in Example 3.8, we use only the data for apartments in average location (reference category) and top location and estimate the model

$$\text{rentsqm} = f_1(\text{area}) + f_2(\text{area}) \text{tlocation} + \beta_0 + \beta_1 \text{tlocation} + \varepsilon. \quad (9.10)$$

The function $f_1(\text{area})$ can be interpreted as the nonlinear effect of the living area in average location ($\text{tlocation} = 0$), and the term $f_1(\text{area}) + f_2(\text{area}) + \beta_1$ can be interpreted as the nonlinear effect of the living area in top location ($\text{tlocation} = 1$). The term $f_2(\text{area}) + \beta_1$ represents the varying effect of living area in top location. Figures 9.4 and 9.5 show



the estimated effects, as well as the functions $\hat{f}_1, \hat{f}_2$. The results are based on `remlreg` objects of `BayesX`.

In the left panel of Fig. 9.4, the estimated effects of the living area for average and top location show the extent of interaction. If the functions are parallel, no interaction is present, and the function $f_2(\text{area})$ (Fig. 9.5, right) should then be (almost) zero. In our case, we do find a relatively strong interaction. For apartments with a small living area, the effect of average and top location is nearly the same. The larger the living area, the more the effects drift apart, i.e., the greater the difference in rent per square meter in average and top locations.

The right panel of Fig. 9.4 shows that the effect of top location is at first small, but then strongly increases as living area increases. From 70 sqm living area onward, the effect remains relatively constant, with an increase of rent of approximately 2 Euro. Hence, the effect of living area clearly depends on location, which can be interpreted as an interaction effect between living area and location.

As mentioned, the model was already estimated with strictly parametric terms in Example 3.8 (p. 102). Recall that the interaction effect was modeled in a linear way. In comparison to this parametric model, we find differences for apartments larger than 70 sqm regarding the varying effect of top location. The AIC is 4,656 in the parametric model and 4,658 in the nonparametric model. Thus the parametric model performs equally well or even slightly better than the nonparametric model.

△

In the case that a basis function approach is used for modeling the functions $f_1, \dots, f_q$ and $f_{z_1|x_1}$ (and $f_{z_2|x_2}$), we obtain

$$f_{z_1|x_1}(z_1) = \sum_{l=1}^d \gamma_{int,l} B_l(z_1) \quad \text{and} \quad f_{z_1|x_1}(z_1) \cdot x_1 = \sum_{l=1}^d \gamma_{int,l} B_l(z_1) x_1.$$

The vector of the interaction term $\mathbf{f}_{int} = (f_{z_1|x_1}(z_{11})x_{11}, \dots, f_{z_1|x_1}(z_{n1})x_{n1})'$ can be represented with the vector $\boldsymbol{\gamma}_{int} = (\gamma_{int,1}, \dots, \gamma_{int,d})'$ of the regression coefficients using

$$\mathbf{f}_{int} = \mathbf{Z}_{int} \boldsymbol{\gamma}_{int}.$$

The design matrix now contains the values of the basis functions multiplied with the respective x_1 values, i.e., $\mathbf{Z}_{int}[i, l] = B_l(z_{i1}) x_{i1}$. Thus, we can write model (9.9) in matrix notation in the form of a large linear model

$$\mathbf{y} = \mathbf{Z}_1 \boldsymbol{\gamma}_1 + \dots + \mathbf{Z}_q \boldsymbol{\gamma}_q + \mathbf{Z}_{int} \boldsymbol{\gamma}_{int} + \mathbf{X} \boldsymbol{\beta} + \boldsymbol{\varepsilon}.$$

This also applies if several interaction or varying coefficient terms are included in the model.

The concept also remains the same, if, instead of z_1 , an interaction $f_{geo|x}(s) \cdot x$ between the location variable s and a binary (or multicategorical) variable x is included in geoadditive models. In this case, we speak of *models with spatially varying coefficients* or *geographically weighted regression*.

9.3.2 Interactions Between Two Continuous Covariates

For two continuous covariates z_1 and z_2 , nonparametric interactions can be considered by adding a smooth two-dimensional function $f_{1|2}(z_1, z_2)$ to the predictor. There are principally two alternatives to do so:

- *Two-dimensional function without main effects:* The effect of z_1 and z_2 is modeled by the two-dimensional surface $f_{1|2}(z_1, z_2)$. One-dimensional main effects $f_1(z_1)$ and $f_2(z_2)$ are not included in the model. This results in the model

$$y_i = f_{1|2}(z_1, z_2) + f_3(z_{i3}) + \dots + f_q(z_{iq}) + \eta_i^{lin} + \varepsilon_i,$$

where once again the level of the function $f_{1|2}(z_1, z_2)$ is not identified. Thus, the surface needs to be centered around zero, i.e.,

$$\sum_{i=1}^n f_{1|2}(z_{i1}, z_{i2}) = 0.$$

The interpretation of the results is more involved than in models with pure main effects. In the following Example 9.4 we will show how an appropriate presentation of results can help with interpretation.

Prior to estimating a two-dimensional interaction, we should be sure that we have enough data combinations of z_1 and z_2 ; a simple scatter plot between z_1 and z_2 can be useful to help determine if this is the case. If there are data gaps in subareas, then the results from these areas should be interpreted carefully. It may be possible that the existing data set is not sufficient to estimate a two-dimensional surface between z_1 and z_2 . Then the specification of a pure main effects model may be preferable.

- *Two-dimensional function with main effects:* In this case, we include the two-dimensional function $f_{1|2}(z_1, z_2)$ in addition to the one-dimensional main effects $f_1(z_1)$ and $f_2(z_2)$ into the model. We then obtain

$$y_i = f_1(z_{i1}) + f_2(z_{i2}) + f_{1|2}(z_1, z_2) + f_3(z_{i3}) + \dots + f_q(z_{iq}) + \eta_i^{lin} + \varepsilon_i.$$

With the inclusion of the main effects, the problem of identifiability becomes more complicated. The type of constraints for $f_{1|2}$ depends on the choice of basis functions used for the main effects and the interaction and also depends on penalization. In general, identifiability is guaranteed if (in addition to centering all functions around zero) also all slices of the interaction $f_{1|2}(z_1, z_2)$, i.e., all one-dimensional smooths with fixed value of z_1 or z_2 , are centered around zero. This approach also facilitates interpretation. The interaction $f_{1|2}(z_1, z_2)$ can then be interpreted as a deviation from the main effects. This modeling variant requires an even larger sample size in order to obtain reliable results. Moreover, a meaningful interpretation of the interaction surface requires some experience.

For modeling $f_{1|2}$, we generally use tensor product-P-splines, radial basis functions, or kriging approaches as described in Sects. 8.2.1–8.2.3. As stated there, the vector

$$\mathbf{f}_{1|2} = (f_{1|2}(z_{11}, z_{12}), \dots, f_{1|2}(z_{n1}, z_{n2}))'$$

of function evaluations can be written as

$$\mathbf{f}_{1|2} = \mathbf{Z}_{1|2} \boldsymbol{\gamma}_{1|2},$$

using an appropriately defined design matrix $\mathbf{Z}_{1|2}$ and with corresponding coefficient vector $\boldsymbol{\gamma}_{1|2}$. We again obtain a large linear model

$$\mathbf{y} = \mathbf{Z}_1 \boldsymbol{\gamma}_1 + \dots + \mathbf{Z}_q \boldsymbol{\gamma}_q + \mathbf{Z}_{1|2} \boldsymbol{\gamma}_{1|2} + \mathbf{X} \boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

where the coefficients $\boldsymbol{\gamma}_1, \dots, \boldsymbol{\gamma}_q, \boldsymbol{\gamma}_{1|2}$ need to be estimated while considering the penalty restrictions; refer to Sect. 9.6.

Example 9.4 Munich Rent Index—Interaction Between Living Area and Year of Construction



In the geoadditive model of Example 9.2, we replace the two main effects of the living area and the year of construction with a two-dimensional surface $f_{1|2}(\text{area}, \text{yearc})$ and obtain the model

$$\text{rentsqm} = f_{1|2}(\text{area}, \text{yearc}) + f_{\text{geo}}(\text{district}) + \beta_0 + \beta_1 \text{glocation} + \beta_2 \text{tlocation} + \varepsilon.$$

As in Example 8.6 (p. 500), the two-dimensional function $f_{1|2}$ is modeled using tensor product P-splines. First- and second-order differences were considered for penalization. Estimates were carried out using `remlreg` objects of `BayesX`. With first differences, we obtain an AIC of 7,317, whereas second-order differences yield an AIC of 7,304. In either case, we find that the AIC is lower than what was found in the geoadditive model of Example 9.2 (AIC = 7,336). In the following, we limit our comparison to the models based on second-order difference penalties.

Before we interpret the results, we first have a look at the distribution of the observations across the living area and year of construction, as seen in the left panel of Fig. 9.6. The plot illustrates the potential difficulties when estimating two-dimensional covariate effects. It appears that we do not have enough observations for apartments with a living area smaller than 30 or greater than 110 sqm to make reliable statements. In these regions, poorly interpretable effects would not be surprising. For apartments with a living space between 30 and 110 sqm, the amount of data appears to be sufficient (Fig. 9.6, right). Consequently, we limit the interpretation to this interval. In practice we mainly have the choice between four alternatives: The collection of additional data, the consideration of additional data from previous studies, limiting the validity of the rent index to apartments with a living area between 30 and 110 sqm, or the use of models with less complexity (in particular models with only main effects). Perhaps the last choice is one of our best alternatives.

We limit the presentation of results to the interaction between the living area and year of construction. In comparison to Example 9.2, the spatial effects remained almost unchanged. Figure 9.7 (left) shows the estimated two-dimensional surface of the living area and the year of construction. The interpretation of two-dimensional surfaces can be quite involved. A helpful device is often to plot *slices* of the two-dimensional function. The right panel of Fig. 9.7 shows various slices of the living area for the year of construction in 1918, 1957, 1972, and 1991, respectively. The slices show whether or not an interaction is necessary.

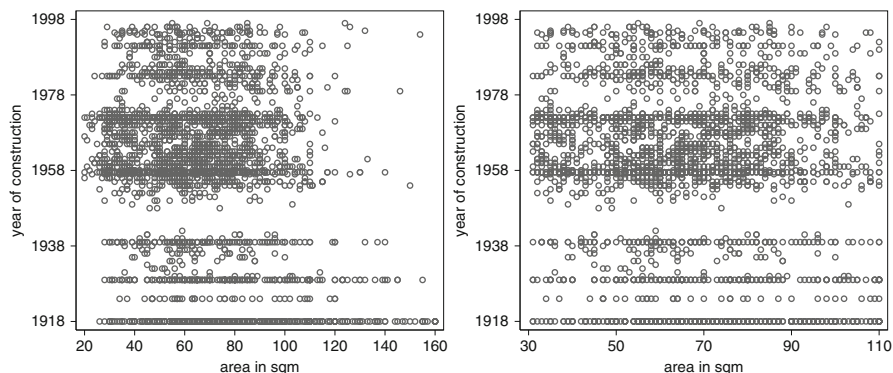


Fig. 9.6 Munich rent index: distribution of area and year of construction. *Left*, complete dataset. *Right*, only observations with area between 30 and 110 square meters

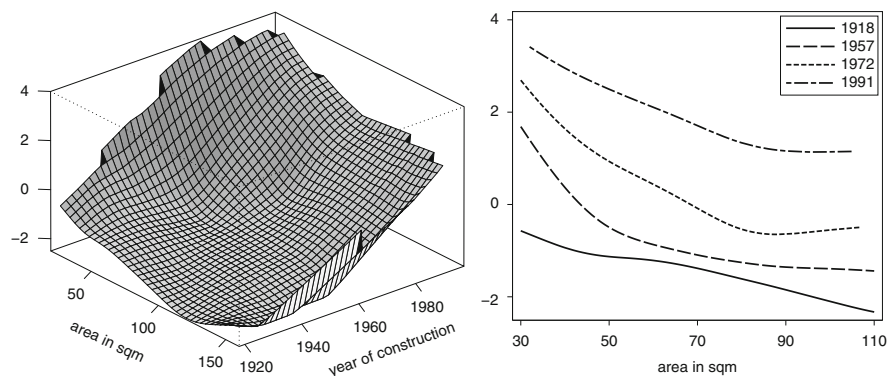


Fig. 9.7 Munich rent index: estimated two-dimensional function $\hat{f}_{1|2}(\text{area}, \text{year})$ (*left*) and slices for various years of construction (*right*)

If the slices run nearly parallel to each other, a noteworthy interaction does not exist, and a model with the two main effects may be sufficient. In our case, the slices do not run parallel to each other and the inclusion of an interaction seems justified. The lower AIC values for the interaction models compared to that of the main effects model also support our conclusion. The slices nicely show that smaller apartments have a larger interaction effect. The decrease of the average rent per sqm is much weaker for older and more modern apartments than for apartments intermediate in age. It is interesting to note that for very old apartments the effect of the living area is almost linear.

△

9.4 Models with Random Effects

In Chap. 7, we discussed random intercept and slope models for longitudinal and cluster data. In this section we show how the semiparametric models considered thus far can be extended in a straightforward way by random coefficient terms. We will

show that such terms have the same structure as nonlinear functions of continuous covariates or spatial effects in a geoaddivitive model.

Suppose we have longitudinal data for individuals $i = 1, \dots, m$ at observed times $t_{i1} < \dots < t_{ij} < \dots < t_{in_i}$ or cluster data for subjects $j = 1, \dots, n_i$ in clusters $i = 1, \dots, m$ as in Chap. 7.

Consider the simple random coefficient model

$$y_{ij} = \gamma_{0i} + \gamma_{1i}x_{ij} + \beta_0 + \beta_1x_{ij} + \varepsilon_{ij}, \quad i = 1, \dots, m, \quad j = 1, \dots, n_i,$$

with fixed intercept β_0 and slope β_1 of covariate x , cluster-specific random intercept γ_{0i} , and random slope γ_{1i} . We restrict ourselves to a priori independent random intercepts $\gamma_{0i} \sim N(0, \tau_0^2)$ and slopes $\gamma_{1i} \sim N(0, \tau_1^2)$. Similarly as in Chap. 7 we express the model in matrix notation in the form

$$\mathbf{y}_i = \mathbf{1}_i \gamma_{0i} + \mathbf{x}_i \gamma_{1i} + \mathbf{1}_i \beta_0 + \mathbf{x}_i \beta_1 + \boldsymbol{\varepsilon}_i,$$

where $\mathbf{y}_i = (y_{i1}, \dots, y_{in_i})'$, $\mathbf{1}_i = (1, \dots, 1)'$ is a n_i -dimensional vector of ones, $\mathbf{x}_i = (x_{i1}, \dots, x_{in_i})'$, $\boldsymbol{\varepsilon}_i = (\varepsilon_{i1}, \dots, \varepsilon_{in_i})'$. Defining $\boldsymbol{\gamma}_0 = (\gamma_{01}, \dots, \gamma_{0m})'$ and $\boldsymbol{\gamma}_1 = (\gamma_{11}, \dots, \gamma_{1m})'$ as the vectors of random intercepts and random slopes, respectively, $\boldsymbol{\beta} = (\beta_0, \beta_1)'$, and the design matrices

$$\mathbf{Z}_0 = \begin{pmatrix} \mathbf{1}_1 & & \mathbf{0} \\ & \ddots & \\ & & \mathbf{1}_i & \\ & & & \ddots \\ & & & & \mathbf{1}_m \end{pmatrix} \quad \mathbf{Z}_1 = \begin{pmatrix} \mathbf{x}_1 & & \mathbf{0} \\ & \ddots & \\ & & \mathbf{x}_i & \\ & & & \ddots \\ & & & & \mathbf{x}_m \end{pmatrix}$$

as well as

$$\mathbf{X} = \begin{pmatrix} \mathbf{1}_1 & \mathbf{x}_1 \\ \vdots & \vdots \\ \mathbf{1}_i & \mathbf{x}_i \\ \vdots & \vdots \\ \mathbf{1}_m & \mathbf{x}_m \end{pmatrix} \quad \mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_i \\ \vdots \\ y_m \end{pmatrix} \quad \boldsymbol{\varepsilon} = \begin{pmatrix} \boldsymbol{\varepsilon}_1 \\ \vdots \\ \boldsymbol{\varepsilon}_i \\ \vdots \\ \boldsymbol{\varepsilon}_m \end{pmatrix},$$

we obtain

$$\mathbf{y} = \mathbf{Z}_0 \boldsymbol{\gamma}_0 + \mathbf{Z}_1 \boldsymbol{\gamma}_1 + \mathbf{X} \boldsymbol{\beta} + \boldsymbol{\varepsilon}.$$

Note the close relationship of random slopes with varying coefficient terms. The effect of x varies with respect to the clusters $i = 1, \dots, m$ as the effect modifier.

As pointed out in Sect. 7.1.1 on p. 354, random coefficient models have an alternative penalized least squares view. Specifically, for fixed variance parameters τ_0^2 and τ_1^2 , the parameters of the model can be estimated by minimizing the penalized least squares criterion

$$PLS(\mathbf{y}_0, \mathbf{y}_1, \boldsymbol{\beta}) = (\mathbf{y} - \mathbf{Z}_0\mathbf{y}_0 - \mathbf{Z}_1\mathbf{y}_1 - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{Z}_0\mathbf{y}_0 - \mathbf{Z}_1\mathbf{y}_1 - \mathbf{X}\boldsymbol{\beta}) \\ + \lambda_0\mathbf{y}'_0\mathbf{K}_0\mathbf{y}_0 + \lambda_1\mathbf{y}'_1\mathbf{K}_1\mathbf{y}_1,$$

where $\mathbf{K}_0 = \mathbf{K}_1 = \mathbf{I}$.

Summarizing, we can express random intercepts and slopes in exactly the same form as nonlinear functions of continuous covariates or the spatial effects in geoadditive models. They can be expressed in the form $\mathbf{Z}\boldsymbol{\gamma}$ with specific penalty $\boldsymbol{\gamma}'\mathbf{K}\boldsymbol{\gamma} = \boldsymbol{\gamma}'\mathbf{I}\boldsymbol{\gamma}$ or in a Bayesian approach with prior $p(\boldsymbol{\gamma} | \tau^2) \propto \exp(-0.5\boldsymbol{\gamma}'\mathbf{K}\boldsymbol{\gamma}/\tau^2) = \exp(-0.5\boldsymbol{\gamma}'\mathbf{I}\boldsymbol{\gamma}/\tau^2)$. In contrast to most other smoothing priors, the random effects priors are not rank deficient.

The next example shows how nonlinear covariate effects and random coefficients can be combined quite naturally in a single regression model with an additive predictor. Such models are often called *additive mixed models*.

Example 9.5 Sales of Orange Juice—Additive Price Effects

This example continues the case study on linear mixed models in Sect. 7.7 (p. 401). There, we estimated the linear mixed model

$$y_{ij} = \beta_0 + \beta_1 \cdot \log p_{ij} + \beta_2 \cdot \log pc1_{ij} + \beta_3 \cdot \log pc2_{ij} + \beta_4 \cdot \log pc3_{ij} \\ + \gamma_{0i} + \gamma_{1i} \cdot \log p_{ij} + \gamma_{2i} \cdot \log pc1_{ij} + \gamma_{3i} \cdot \log pc2_{ij} + \gamma_{4i} \cdot \log pc3_{ij} + \varepsilon_{ij} \quad (9.11)$$

with fixed effects for the (centered) log own-item price and the log-cross prices of competitors, a store-specific random intercept γ_{1i} and random slopes $\gamma_{2i} - \gamma_{4i}$ of the log prices for the $m = 81$ stores, $i = 1, \dots, 81$.

A somewhat unsatisfactory aspect of this model is that the type of nonlinearity of the price effects is determined in advance using a log-transformation of prices. To investigate whether the log-transformation is really appropriate, we can replace the linear fixed effects in Eq. (9.11) by P-splines leading to the model

$$y_{ij} = f_1(\log p_{ij}) + f_2(\log pc1_{ij}) + f_3(\log pc2_{ij}) + f_4(\log pc3_{ij}) \\ + \gamma_{0i} + \gamma_{1i} \log p_{ij} + \gamma_{2i} \log pc1_{ij} + \gamma_{3i} \log pc2_{ij} + \gamma_{4i} \log pc3_{ij} + \beta_0 + \varepsilon_{ij}. \quad (9.12)$$

This is a model that combines nonlinear covariate effects with random coefficients. Since models with nonlinear covariate effects and additional random effects are well suited for a Bayesian approach using MCMC for inference, we use a fully Bayesian approach and `bayesreg` objects of `BayesX` for estimation. See Sects. 8.1.9 (p. 486) and 9.6.3 below for details. We assume a priori a monotonically decreasing effect of the own price and monotonically increasing effects of cross prices. Imposing monotonicity constraints is relatively straightforward in a Bayesian approach; see [Brezger and Steiner \(2008\)](#) for



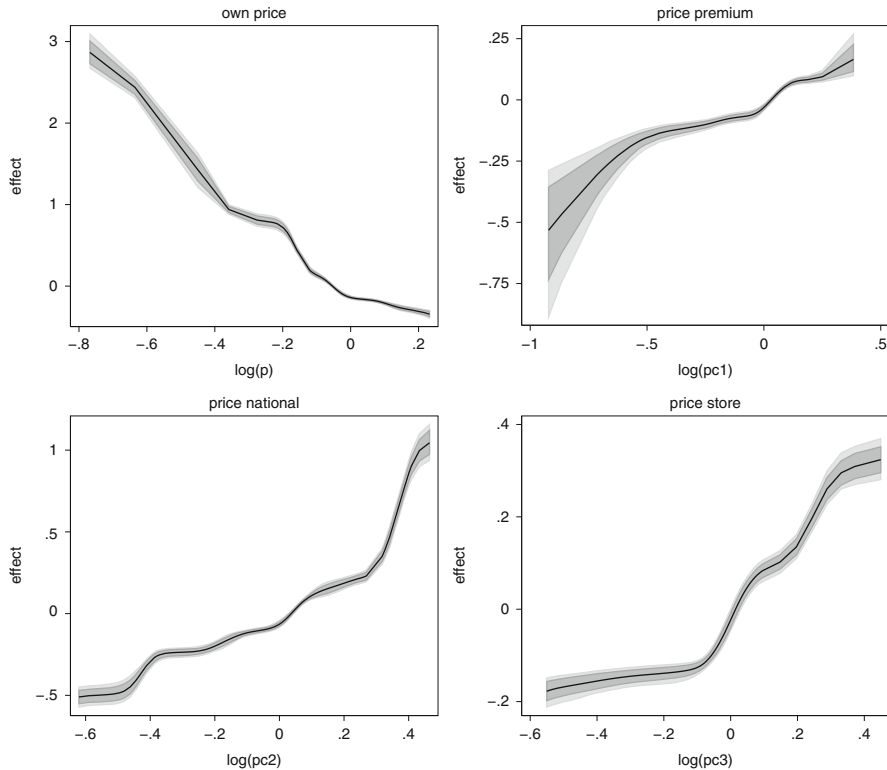


Fig. 9.8 Sales of orange juice: estimated price effects in the model (9.12). Shown is the posterior mean together with 80 % and 95 % pointwise credible intervals

details. If the log-transformation is sufficient to capture the nonlinearity, the estimated curves $\hat{f}_1, \dots, \hat{f}_4$ should be approximately linear.

In matrix notation model (9.12) can be expressed in additive form as

$$\mathbf{y} = \mathbf{Z}_1\boldsymbol{\gamma}_1 + \dots + \mathbf{Z}_9\boldsymbol{\gamma}_9 + \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

where $\mathbf{Z}_1\boldsymbol{\gamma}_1$ to $\mathbf{Z}_4\boldsymbol{\gamma}_4$ correspond to the P-spline smooth terms of the own-item price and cross prices, $\mathbf{Z}_5\boldsymbol{\gamma}_5$ corresponds to the random intercept, $\mathbf{Z}_6\boldsymbol{\gamma}_6$ to $\mathbf{Z}_9\boldsymbol{\gamma}_9$ correspond to the random slopes of the price effects, and $\mathbf{X}\boldsymbol{\beta}$ corresponds to the fixed effects. The matrices $\mathbf{Z}_1$ to $\mathbf{Z}_4$ are the standard P-spline design matrices, as outlined in Sect. 8.1.2.

Figure 9.8 shows the estimated price effects. Although considerable additional nonlinearity remains, the plots reveal that the log-transformation is a reasonable approximation to the nonlinear effects. The size of the random effects measured through the random effects variances is in the same range as in the parametric model of Sect. 7.7.

To demonstrate the outlet-specific heterogeneity, Fig. 9.9 shows some outlet-specific log-sales own-price curves which are now additively composed of the random intercept, the nonlinear log-price effect f_1 , and the linear log-price random effect $\gamma_{1i}\log p_{ij}$.

△

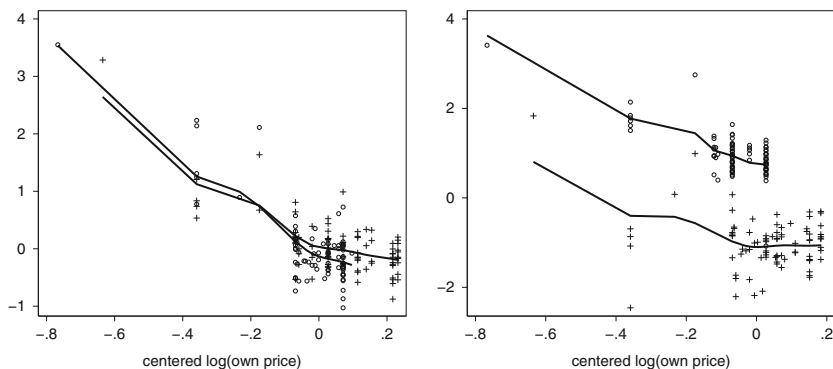


Fig. 9.9 Sales of orange juice: log-sales versus (*centered*) log-price curves in model (9.12) for four different outlets, together with corresponding partial residuals

9.5 Structured Additive Regression

The various terms within the model classes considered thus far can be combined arbitrarily. For example, a particular model could include all main effects, a spatial effect, and one or more terms with varying coefficients for modeling interactions. As shown in the previous section, it is also possible that additional random effects can be included as in mixed models. The resulting models can be described in a unified and general form as *structured additive regression (STAR) models*,

$$y = f_1(v_1) + \dots + f_q(v_q) + \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + \varepsilon, \quad (9.13)$$

where $v_1, \dots, v_q$ are one- or multidimensional covariates of different types constructed from the original variables. The functions can also be of different structure. Specific types of covariates and functions yield the following special cases:

$$\begin{aligned} f_1(v_1) &= f_1(z_1), & v_1 &= z_1, & \text{nonlinear effect of } z_1, \\ f_2(v_2) &= f_{geo}(s), & v_2 &= s, & \text{spatial effect of location variable } s, \\ f_3(v_3) &= f(z)x, & v_3 &= (z, x), & \text{varying effect of } x \text{ with } z, \\ f_4(v_4) &= f_{1|2}(z_1, z_2), & v_4 &= (z_1, z_2), & \text{nonlinear interaction between } z_1 \text{ and } z_2, \\ f_5(v_5) &= \gamma_i, & v_5 &= i & \text{individual-specific random intercept,} \\ f_6(v_6) &= \gamma_i u, & v_6 &= (u, i), & \text{individual-specific random slope of } u. \end{aligned}$$

Recall that it is also possible to express the linear part $\beta_0 + \beta_1 x_1 + \dots + \beta_k x_k = \mathbf{x}'\boldsymbol{\beta}$ of the predictor as the function $f_0(v_0) = \mathbf{x}'\boldsymbol{\beta}$, where $v_0 = (1, x_1, \dots, x_k)'$. Since

Table 9.2 Overview of commonly used model terms in STAR models

Term type	Design matrix V	Penalty matrix K	Section
P-spline	Basis functions evaluated at the observations	$K = D_r' D_r$, with D_r a r th order difference matrix	8.1.2
General basis functions approach	Basis functions evaluated at the observations	Based on the integral of squared second derivatives	8.1.3
Varying coefficient term	Basis functions multiplied with the interaction variable	$K = D_r' D_r$ as for P-splines	9.3.1
2D-P-spline	2D-basis functions evaluated at the observations	$K = I \otimes K_1 + K_2 \otimes I$ with identity matrix I and penalty matrices K_1 and K_2 as for univariate P-splines	8.2.1
Kriging	Basis functions based on the correlation function	$K = R$, with correlation matrix R	8.1.6, 8.2.3
Markov random field	0/1-incidence matrix, that links observations and regions	$K =$ neighborhood matrix	8.2.4
Random intercept	0/1-incidence matrix, that links observations and clusters	$K = I$, with identity matrix I	Chap. 7, 9.4

the linear part, or at least the constant β_0 , is always part of the model, it is explicitly listed in Eq. (9.13).

If the functions $f_1, \dots, f_q$ are modeled and estimated with basis functions, we obtain a large linear model, which is in the spirit of Eilers and Marx (2002). Using the vectors $f_1, \dots, f_q$ of functions evaluated at the observations of the covariates $v_1, \dots, v_q$, we obtain

$$y = f_1 + \dots + f_q + X\beta + \varepsilon = V_1\gamma_1 + \dots + V_q\gamma_q + X\beta + \varepsilon,$$

with appropriately defined design matrixes V_j and coefficient vectors γ_j . The smoothness of the functions will be regularized with penalties of the form $\lambda_j \gamma_j' K_j \gamma_j$, with smoothing parameter λ_j and penalty matrix K_j . In a fully Bayesian approach, each penalty corresponds to a smoothness prior of the form

$$p(\gamma_j | \tau_j^2) \propto \left(\frac{1}{\tau_j^2} \right)^{\text{rk}(K_j)/2} \exp \left(-\frac{1}{2\tau_j^2} \gamma_j' K_j \gamma_j \right).$$

Thus every model term in STAR models is characterized by the form of the design matrix V_j and the penalty matrix K_j . Table 9.2 gives an overview of commonly used terms. Details of parameter estimation will be provided in the next section.

The unified concept of STAR models can also be extended to non-normal regression settings, in particular binary, discrete, and categorical responses. As with generalized linear models, we assume that the response variables y_i are (conditionally) independent given the predictor η_i . The structured additive predictor

$$\eta_i^{struct} = f(v_{i1}) + \dots + f(v_{iq}) + \mathbf{x}_i' \boldsymbol{\beta}$$

is linked to the (conditional) mean $E(y_i) = \mu_i$ through

$$E(y_i) = \mu_i = h(\eta_i^{struct}),$$

with the response function h . We refer to the resulting model class as *generalized STAR models*.

This generalized model class comprises all model classes discussed so far as special cases. For example, choosing

$$\eta_i^{struct} = \eta_i^{add} = f_1(z_{i1}) + \dots + f_q(z_{iq}) + \mathbf{x}_i' \boldsymbol{\beta}$$

results in a *generalized additive model* (GAM). With

$$\eta_i^{struct} = \eta_i^{add} + f_{geo}(s_i),$$

we obtain *generalized geoadditive models*, and so on.

Example 9.6 Vehicle Insurance—Claim Frequency

We have already discussed examples of generalized STAR models in Chap. 2, as seen in the Examples 2.12, 2.13, and 2.15. In Example 2.15 (p. 57), we modeled the claim frequency of a vehicle insurance policy using a *structured additive Poisson model* with the predictor

$$\eta^{struct} = f_1(\text{age}) + f_2(\text{age.v}) + f_3(\text{hp}) + f_4(\text{bm}) + f_5(\text{district}) + \beta_0 + \beta_1 \text{gender} + \dots$$

In a case study, [Denuit and Lang \(2005\)](#) modeled an additional interaction between gender and policyholder's age in form of an age-varying gender effect. This yields the extended structured additive predictor

$$\begin{aligned} \eta^{struct} = & f_1(\text{age}) + f_2(\text{age.v}) + f_3(\text{hp}) + f_4(\text{bm}) + f_5(\text{district}) \\ & + f_6(\text{age, gender}) + \beta_0 + \beta_1 \text{gender} + \dots, \end{aligned}$$

where the interaction f_6 has the form $f_6(\text{age, gender}) = f(\text{age}) \cdot \text{gender}$ of a varying coefficient term with a nonparametric function f of age . All nonlinear functions are modeled as P-splines with 20 interior knots and second-order difference penalty. The district effect is based on a Markov random field. All results are based on mixed model technology as will be outlined in Sect. 9.6.2. We used `relmreg` objects of BayesX for estimation.

Figure 9.10 (left panel) shows the estimated age effect $\hat{f}_1$ for women and the age effect $\hat{f}_1 + \hat{f}_6 + \hat{\beta}_1$ for men ($\text{gender} = 1$). The right panel of Fig. 9.10 displays the effect $\hat{f}_6 + \hat{\beta}_1$ of men compared to women varying with respect to age .

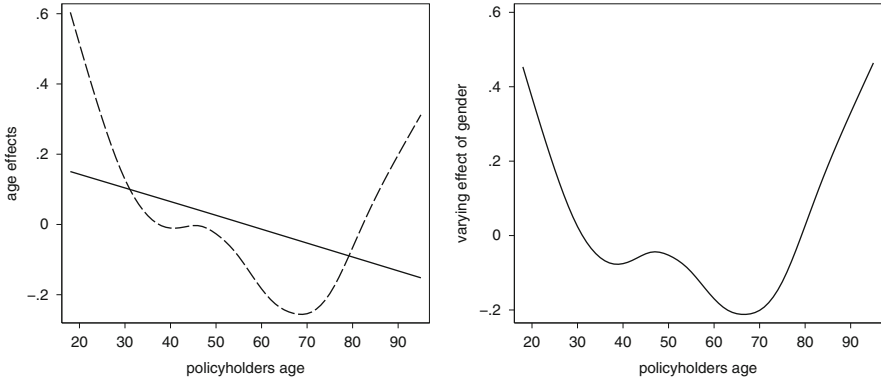


Fig. 9.10 Vehicle insurance: The *left panel* shows the effect of the policyholders age for females (*solid line*) and males (*dashed line*). The *right panel* shows the effect of males in comparison to females varying with respect to age

As women get older, the average damage frequency decreases linearly. For men we obtain the nonlinear form that was already discussed in the examples of Chap. 2. Younger and very old men show a strongly increased tendency towards having accidents. Men between the ages of 30 and 80 years cause less damage than women of the same age group. Overall, the interaction effect is quite pronounced. Further discussion of the other results do not differ much from those outlined in Chap. 2.

△

In the next example we will show how to extend the STAR approach to structured additive categorical regression models. The example also shows that care has to be taken when modeling spatial effects, especially if some covariates (e.g., in ecological applications) are *location-specific*. It is then possible that the spatial heterogeneity effect and the influence of these variables cannot be separated. Problems of this kind have only been recently resolved in the literature. See [Lawson and Liu \(2007\)](#), [Hodges and Reich \(2010\)](#), and [Paciorek \(2010\)](#) for various investigations and possible solutions.

Example 9.7 Forest Health Status—Ordinal Logit Model



We revisit the Examples 1.4 (p. 9) and 6.5 (p. 337) of Chaps. 1 and 6 and extend the linear predictors $\eta_{itr} = \theta_{0r} + \mathbf{x}'_{it}\boldsymbol{\beta}$, $r = 1, 2$ of the three-categorical ordinal logit model $P(Y_{it} = r)$ for tree i in year t to the additive predictors

$$\eta_{itr} = \theta_{0r} + f_1(\text{age}_{it}) + f_2(\text{gradient}_i) + f_3(\text{canopyd}_{it}) + f_4(t) + f_5(\text{alt}_i) + \mathbf{x}'_{it}\boldsymbol{\beta}.$$

In comparison to Example 6.5, the parametric effects of most of the continuous covariates are replaced by possibly nonlinear functions. Only the effect of soil depth (*depth*) and pH-value (*ph*) are still modeled linearly, since previous nonparametric analyses resulted in linear effects for these variables. The nonlinear effects f_1 through f_5 are based on cubic P-splines with 20 interior knots and second-order difference penalties. The covariate vector $\mathbf{x}_{it}$ contains the remaining regressors with linear effects $\boldsymbol{\beta}$, as listed in Table 9.3. Figure 9.11 contains the estimated nonlinear effects f_1 through f_5 (estimated using a mixed

9.3 Generalized STAR Models

Model

The response variables are (conditionally) independent and the (conditional) density of y_i belongs to an exponential family. The (conditional) mean μ_i is connected to the structured additive predictor

$$\eta_i^{struct} = f(v_{i1}) + \dots + f(v_{iq}) + \mathbf{x}_i' \boldsymbol{\beta}$$

through

$$\mu_i = h(\eta_i^{struct}) \quad \text{or} \quad \eta_i^{struct} = g(\mu_i).$$

The $v_1, \dots, v_q$ are one- or multidimensional covariates of differing types constructed from the original variables. See Table 9.2 for an overview of the various model terms.

Software

- R package `mgcv`: See Wood (2006) for details. The package provides various uni- and multidimensional smoothers to estimate univariate STAR models. Smoothing parameter estimation is based on minimizing GCV or REML estimation. A similar class of models is available in the package `amer`.
- Software package `BayesX` and R package `R2BayesX`: Estimation is carried out either using mixed model methodology (`remlreg` objects), fully Bayesian inference based on MCMC (`bayesreg` and `mcmcreg` objects), or penalized likelihood in combination with variable selection techniques for smoothing parameter selection and model choice (`stepwisereg` objects). The software also provides methodology for categorical responses.
- R package `gamlss`: The package even allows the specification of additive predictors for other parameters associated with the response distribution (e.g., the mean and the variance for Gaussian responses). The software is limited to pure additive predictors.
- R packages: `mboost` and `gamboostLSS` for estimating STAR models with boosting.

model representation, see Sect. 9.6.2). All results are based on `relmreg` objects of the software `BayesX`.

We obtain an AIC of 1,682.8, which indicates a clear improvement in goodness of fit compared to the parametric model with an AIC of 2,029.16 (see Example 6.5 on p. 337).

The covariate effects do, however, show some surprising results:

- As in the parametric model, younger trees appear to be less damaged than older trees. However, we observe a decreasing effect for very old trees which contradicts our expectations on a monotonically increasing relationship.

Table 9.3 Forest health status: estimated linear effects in the model without spatial effects (left) and in the model with spatial effects (right)

	Coefficient	p-value	Coefficient	p-value
θ_1	−4.000	0.009	−1.515	0.453
θ_2	1.107	0.470	4.846	0.019
<i>depth</i>	−0.048	<0.001	−0.006	0.838
<i>ph</i>	−0.718	0.018	−0.249	0.491
<i>type</i>	−0.644	<0.001	−0.202	0.373
<i>fert</i>	−0.571	0.004	−0.570	0.253
<i>humus0</i>	−0.419	0.008	−0.453	0.014
<i>humus2</i>	0.413	0.002	0.279	0.057
<i>humus3</i>	0.087	0.573	0.218	0.223
<i>humus4</i>	−0.032	0.879	0.192	0.443
<i>watermoisture1</i>	−0.801	<0.001	−0.762	0.167
<i>watermoisture3</i>	0.483	0.001	0.812	0.031
<i>alkali1</i>	1.165	<0.001	0.920	0.144
<i>alkali3</i>	−0.966	<0.001	−0.359	0.523
<i>alkali4</i>	0.195	0.507	−0.201	0.798

- The effect of the gradient of slope is extremely rough and not interpretable.
- Trees in a lower and higher altitude appear to be less damaged than trees in moderate altitude. This effect is somewhat unexpected, as we might expect a monotonically increasing effect.
- Compared to the results obtained from Example 6.5, the effects of the linear covariates are relatively unstable; see, for example, the effect of the alkalinity.

How can we explain these somewhat peculiar results? As a first step, we extend the regression model by an additional spatial effect using a two-dimensional P-spline. Table 9.3 again shows the resulting linear effects. Figure 9.12 shows the resulting nonlinear effects, which dramatically change with the inclusion of the spatial effect. Figure 9.13 displays the spatial effect $f_{geo}(s)$. We observe the following:

- The decrease of the age effect for very old trees disappears.
- The effects of soil depth (*depth*), pH-value (*ph*), type of forest (*type*), alkalinity (*alkali*), gradient (*gradient*), and altitude (*alt*) either disappear, or are at least considerably reduced. All these variables are location-specific covariates whose value either does not change at all or very little over time.

In particular, the last observation provides essential information towards explaining the surprising results found with our initial estimates. In the first model without spatial effects, spatial heterogeneity was entirely captured by location-specific covariates. However, the location-specific effects appear to be rather complicated in this application. Apparently, the gradient and the altitude do absorb the effect of further, unobserved or unrecognized, covariates. As a result, their effects are difficult to interpret as they are confounded with other effects. In the second model, the flexibly modeled spatial heterogeneity effect covers the effects of the location-specific covariates and to a large extent absorbs their influences. However, this is not desirable, since the spatial effect is included to absorb the effects of unobserved covariates, but not the effects of observed covariates.

Which conclusion can we draw from this? It seems that in this application it is not possible or at least very difficult to separate the different effects. A possible solution

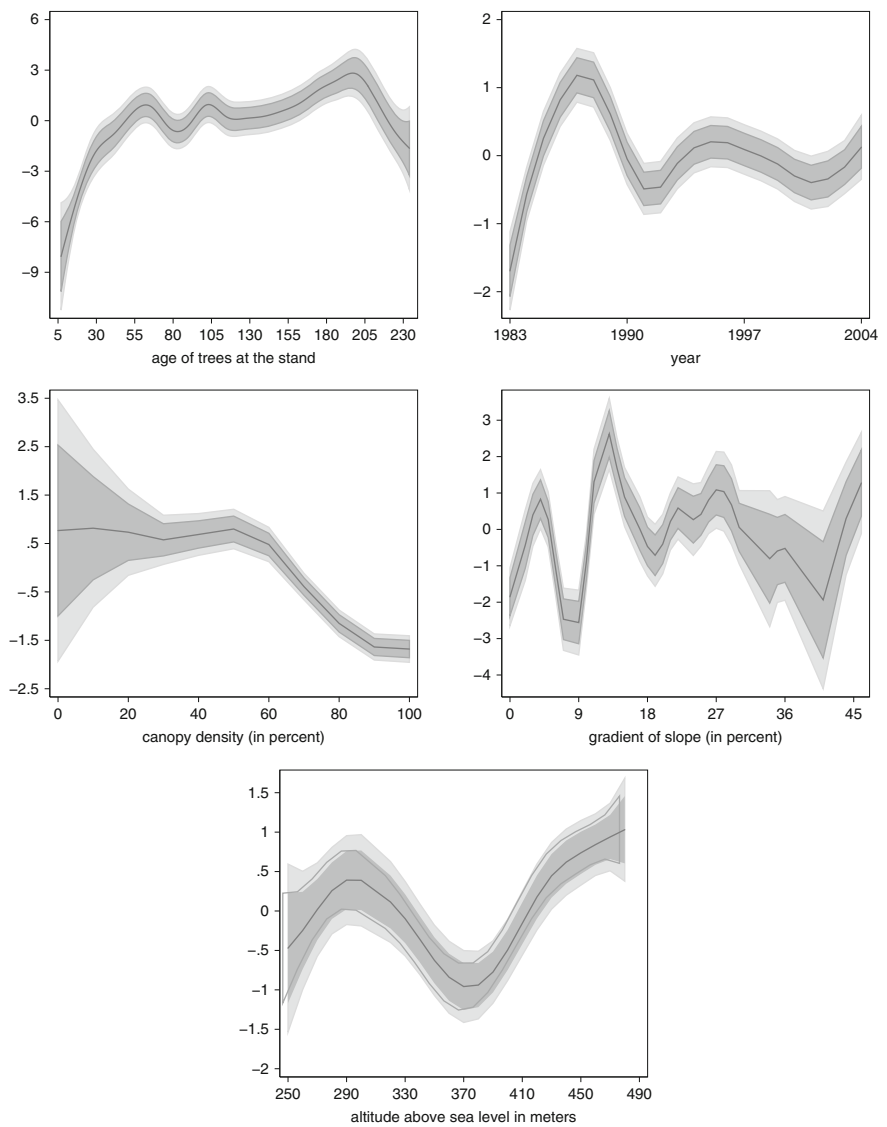


Fig. 9.11 Forest health status: estimated nonlinear effects in the model excluding spatial effects. Additionally included are pointwise 80 % and 95 % confidence intervals

would be the estimation of a model with spatial effects, but without location-specific covariates. By carefully examining the properties of the locations with especially large (or small) spatial effects, we could then try to extract the relevant location-specific covariates. The definition of new covariates, e.g., well-chosen linear combinations, could be useful. Again we defer to the review of recent literature on spatial confounding provided above.

△

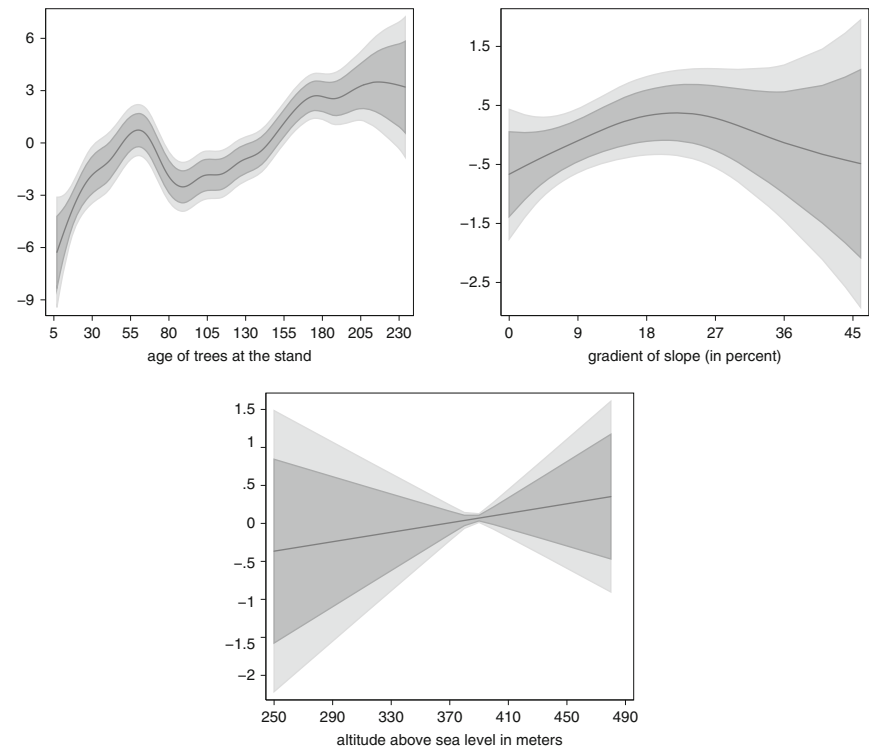
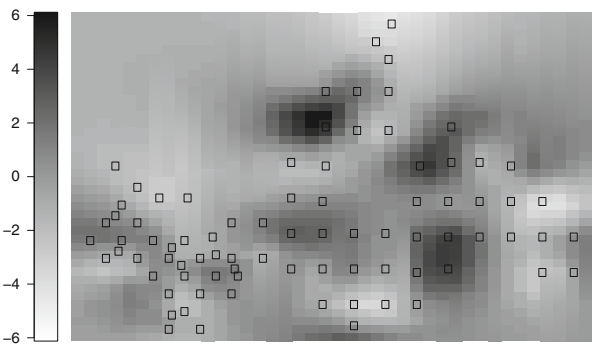


Fig. 9.12 Forest health status: selected nonlinear effects in the model with spatial effect included, along with pointwise 80 % and 95 % confidence intervals

Fig. 9.13 Forest health status: estimated spatial effect



9.6 Inference

We restrict ourselves to the estimation of STAR models that are based on basis functions approaches. We will consider four different inferential concepts:

- Penalized least squares or likelihood estimates in combination with the minimization of a model choice criterion for the choice of smoothing parameters (Sect. 9.6.1)
- Inference based on a mixed model representation (Sect. 9.6.2)
- Fully Bayesian inference based on MCMC simulation techniques (Sect. 9.6.3)
- Boosting for structured additive regression models (Sect. 9.7)

9.6.1 Penalized Least Squares or Likelihood Estimation

We first consider STAR models

$$\mathbf{y} = \mathbf{f}_1 + \dots + \mathbf{f}_q + \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} = \mathbf{V}_1\boldsymbol{\gamma}_1 + \dots + \mathbf{V}_q\boldsymbol{\gamma}_q + \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \quad (9.14)$$

with a continuous response variable y . Extending the penalized least squares criterion of Chap. 8, we minimize the criterion

$$\text{LS}_{pen}(\boldsymbol{\gamma}_1, \dots, \boldsymbol{\gamma}_q, \boldsymbol{\beta}) = \text{LS}(\boldsymbol{\gamma}_1, \dots, \boldsymbol{\gamma}_q, \boldsymbol{\beta}) + \sum_{j=1}^q \lambda_j \boldsymbol{\gamma}_j' \mathbf{K}_j \boldsymbol{\gamma}_j \quad (9.15)$$

with respect to $\boldsymbol{\gamma}_1, \dots, \boldsymbol{\gamma}_q, \boldsymbol{\beta}$. In this setting, the least squares criterion is defined as

$$\text{LS}(\boldsymbol{\gamma}_1, \dots, \boldsymbol{\gamma}_q, \boldsymbol{\beta}) = (\mathbf{y} - \mathbf{V}_1\boldsymbol{\gamma}_1 - \dots - \mathbf{V}_q\boldsymbol{\gamma}_q - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{V}_1\boldsymbol{\gamma}_1 - \dots - \mathbf{V}_q\boldsymbol{\gamma}_q - \mathbf{X}\boldsymbol{\beta}).$$

As in Chap. 8, the form of the penalty matrix $\mathbf{K}_j$ depends on the specific type of f_j and also on the chosen penalty. An overview is provided in Table 9.2. The smoothing parameter λ_j governs the trade-off between fidelity to the data and smoothness controlled by the penalty $\boldsymbol{\gamma}_j' \mathbf{K}_j \boldsymbol{\gamma}_j$. We first assume that λ_j is known.

To minimize Eq. (9.15), we can choose between two alternatives: iterative minimization using the backfitting algorithm or direct minimization of the penalized least squares criterion. The backfitting algorithm has the advantage that estimation of complex models can be traced back to one- or two-dimensional smoothing as described in Chap. 8. In addition, the backfitting algorithm allows any combination of smoothers as building blocks for the estimation of STAR models. This implies that basis function approaches as well as local smoothers can be used as building blocks or combinations thereof.

9.4 Backfitting Algorithm

1. Initialization: set, for example, $\hat{f}_1 = \dots = \hat{f}_q = \mathbf{0}, \hat{\beta} = \mathbf{0}$.
2. Obtain updated estimates $\hat{f}_j, j = 1, \dots, q$, through

$$\hat{f}_j = S_j(y - \sum_{l \neq j} \hat{f}_l - X\hat{\beta}).$$

3. Obtain updated estimates $\hat{\beta}$, through

$$\hat{\beta} = (X'X)^{-1}X'(y - \hat{f}_1 - \dots - \hat{f}_q).$$

4. Iterate the steps 2–3 until the estimated functions do not deviate more than a small given increment in two subsequent iterations.

Backfitting

The idea of backfitting is to obtain estimates $\hat{f}_1, \dots, \hat{f}_q, \hat{\beta}$ iteratively by the smoothing of partial residuals. Neglecting the errors ε in Eq. (9.14), the following approximately holds:

$$f_j \approx y - f_1 - \dots - f_{j-1} - f_{j+1} - \dots - f_q - X\beta.$$

In the case that we already have estimates for $\hat{\beta}$ and $\hat{f}_l, l \neq j$, we can view $y - \sum_{l \neq j} \hat{f}_l - X\hat{\beta}$ as a (partial) residual vector (without f_j). In order to estimate f_j , we apply the scatter plot or bivariate smoother S_j associated with f_j to this residual vector. We obtain

$$\hat{f}_j = S_j(y - \sum_{l \neq j} \hat{f}_l - X\hat{\beta}).$$

All estimators that are based on basis function approaches have the form

$$\hat{f}_j = V_j(V_j'V_j + \lambda_j K_j)^{-1}V_j'(y - \sum_{l \neq j} \hat{f}_l - X\hat{\beta}).$$

The particular form of V_j and K_j depends on the type of smoother used, as outlined in Table 9.2. Given starting values for $\hat{\beta}, \hat{f}_1, \dots, \hat{f}_q$, this concept is applied iteratively and results in the *backfitting algorithm* of Box 9.4.

Note that we have deliberately ignored the dependence of the smoothers S_j on the unknown smoothing parameters. The choice of the smoothing parameters will be further discussed below.

Direct Minimization of the Penalized Least Squares Criterion

In many cases, it is possible to compute the penalized least squares estimator through direct minimization of Eq.(9.15) with a non-iterative procedure. For instance the `mgcv` package of R works with direct optimization. Minimization can be achieved by computing the first derivatives of the criterion with respect to the unknown parameters, setting the derivatives to zero and finally solving the resulting system of equations.

Setting the first derivatives to zero results in the system of equations

$$\begin{pmatrix} V_1' V_1 + \lambda_1 K_1 & \cdots & V_1' V_q & V_1' X \\ \vdots & \ddots & \vdots & \vdots \\ V_q' V_1 & \cdots & V_q' V_q + \lambda_q K_q & V_q' X \\ X' V_1 & \cdots & X' V_q & X' X \end{pmatrix} \begin{pmatrix} \gamma_1 \\ \vdots \\ \gamma_q \\ \beta \end{pmatrix} = \begin{pmatrix} V_1' y \\ \vdots \\ V_q' y \\ X' y \end{pmatrix}. \quad (9.16)$$

The direct (non-iterative) solution of the system of equations works well with a comparably small number of regression coefficients, e.g., if q is small and the functions $f_1, \dots, f_q$ are specified by P-splines having a relatively modest number of knots. For high-dimensional $\gamma_1, \dots, \gamma_q$, the numerical solution of the system of Eq.(9.16) can be obtained through the iterative *Gauß–Seidel algorithm*, which in turn is identical to the backfitting algorithm described above; see Buja, Hastie, and Tibshirani (1989).

Similar to linear models, we can estimate the (conditional) mean $E(y)$ of the response by $\hat{y} = H y$ using the estimated regression coefficients. The hat (prediction) matrix or smoother matrix H is given by

$$H = (V_1 \ V_2 \ \dots \ V_q \ X) \begin{pmatrix} V_1' V_1 + \lambda_1 K_1 & \cdots & V_1' V_q & V_1' X \\ \vdots & \ddots & \vdots & \vdots \\ V_q' V_1 & \cdots & V_q' V_q + \lambda_q K_q & V_q' X \\ X' V_1 & \cdots & X' V_q & X' X \end{pmatrix}^{-1} \begin{pmatrix} V_1' \\ \vdots \\ V_q' \\ X' \end{pmatrix}.$$

Generalized STAR Models

For generalized structured additive models with non-normally distributed response and the predictor $\eta^{struct} = V_1 \gamma_1 + \dots + V_q \gamma_q + X \beta$, estimation is achieved by maximizing the penalized log-likelihood criterion

$$l_{pen}(\gamma_1, \dots, \gamma_q, \beta) = l(\gamma_1, \dots, \gamma_q, \beta) - \frac{1}{2} \sum_{j=1}^q \lambda_j \gamma_j' K_j \gamma_j$$

with respect to $\gamma_1, \dots, \gamma_q, \beta$. Thereby $l(\gamma_1, \dots, \gamma_q, \beta)$ denotes the usual log-likelihood of generalized linear models but with the linear predictor η^{lin} replaced by η^{struct} . Setting the first derivatives to zero leads to the high-dimensional system of equations

$$s_{pen,0}(\boldsymbol{y}_1, \dots, \boldsymbol{y}_q, \boldsymbol{\beta}) = s_0(\boldsymbol{y}_1, \dots, \boldsymbol{y}_q, \boldsymbol{\beta}) = \mathbf{0}$$

$$s_{pen,j}(\boldsymbol{y}_1, \dots, \boldsymbol{y}_q, \boldsymbol{\beta}) = s_j(\boldsymbol{y}_1, \dots, \boldsymbol{y}_q, \boldsymbol{\beta}) - \lambda_j \boldsymbol{K}_j \boldsymbol{y}_j = \mathbf{0}, \quad j = 1, \dots, q.$$

The functions $s_0(\boldsymbol{y}_1, \dots, \boldsymbol{y}_q, \boldsymbol{\beta}) = \partial l / \partial \boldsymbol{\beta}$, $s_j(\boldsymbol{y}_1, \dots, \boldsymbol{y}_q, \boldsymbol{\beta}) = \partial l / \partial \boldsymbol{y}_j$ have the form of the score function in generalized linear models. The structure of the system of equations is similar to that of Eq. (9.16). However, similar to generalized linear models, the equations depend on working weights and working observations. As presented in Chap. 5, the system of equations can be solved with the Fisher scoring algorithm or in combination with backfitting; see, for example, [Fahrmeir and Tutz \(2001, Chap. 5\)](#), [Tutz \(2011, Chap. 8\)](#), [Eilers and Marx \(2002\)](#), and [Marx and Eilers \(1998\)](#).

Choice of the Smoothing Parameters

The smoothing parameters are estimated by optimizing an appropriate model choice criterion, e.g., AIC or GCV.

To minimize AIC, we have to take the effective dimension of the model or effective number of parameters into consideration. The effective number of parameters can be defined in a generalized way, as outlined in Sect. 8.1.8 (p. 473), with the trace of the full STAR model smoother matrix $\boldsymbol{H}$. For Gaussian responses, the smoother matrix projects the vector $\boldsymbol{y}$ into $\hat{\boldsymbol{y}} = \boldsymbol{H} \boldsymbol{y}$. For non-Gaussian response, we use the corresponding matrix from the working model. For STAR models with Gaussian errors, the AIC criterion is then given by

$$\text{AIC} = n \log(\hat{\sigma}^2) + 2(\text{tr}(\boldsymbol{H}) + 1),$$

where $\hat{\sigma}^2 = \sum (y_i - \hat{\eta}_i^{\text{struct}})^2 / n$ is the ML estimator of the error variance.

For the generalized cross validation score GCV, we obtain

$$\text{GCV} = \frac{1}{n} \sum_{i=1}^n \left(\frac{y_i - \hat{\eta}_i^{\text{struct}}}{1 - \text{tr}(\boldsymbol{H})/n} \right)^2.$$

If we use the iterative backfitting algorithm to estimate the regression coefficients, we do not obtain the full smoother matrix and (exact) computation of the criteria is, thus, not possible. In practice, the trace of the full smoother matrix is often approximated via the sum of the traces of individual smoother matrices. Provided that the correlation among individual terms is not too strong, this approximation works well in practice.

Until approximately the year 2000, a simple but not very effective grid search was widely used to minimize AIC or GCV, and thus problems with more than, say four or five smoothing parameters, were barely manageable. Currently, much more efficient (albeit technically complex) algorithms for the optimization of model choice criteria are available. The monograph by [Wood \(2006\)](#) offers a detailed description of the

9.5 Residuals in STAR Models with Continuous Response

Ordinary Residuals

$$\hat{\varepsilon}_i = y_i - \hat{\eta}_i^{struct} = y_i - \hat{f}_1(v_{i1}) - \dots - \hat{f}_q(v_{iq}) - \mathbf{x}_i' \hat{\boldsymbol{\beta}}.$$

Standardized Residuals

The i th standardized residual is given by

$$r_i = \frac{y_i - \hat{\eta}_i^{struct}}{\hat{\sigma} \sqrt{1 - h_{ii}}},$$

where h_{ii} is the i th diagonal element of the full smoother matrix $\mathbf{H}$.

Partial Residuals

Partial residuals regarding v_j are defined by

$$\begin{aligned} \hat{\varepsilon}_{v_j, i} &= y_i - \hat{f}_1(v_{i1}) - \dots - \hat{f}_{j-1}(v_{i,j-1}) - \hat{f}_{j+1}(v_{i,j+1}) - \dots - \hat{f}_q(v_{iq}) - \mathbf{x}_i' \hat{\boldsymbol{\beta}} \\ &= y_i - \hat{\eta}_i^{struct} + \hat{f}_j(v_{ij}). \end{aligned}$$

methodological background; see also [Wood \(2000, 2004, 2008\)](#). An implementation can be found in the R package `mgcv`.

Model Choice and Diagnosis

Relatively little literature exists about model choice and diagnosis for nonparametric and semiparametric models, and user-friendly software is even rarer. However, some of the concepts regarding model choice in linear models, discussed in Sect. 3.4, easily carry over to STAR models. For instance, competing models can be compared via standard model choice criteria such as AIC or GCV, even though there are some restrictions to be considered when the model has been estimated using the mixed model representation of penalized regression; see the next section.

Model diagnostics centrally relies on the various types of residuals. Ordinary, standardized, and partial residuals can be defined analogously to those of the linear model. Refer to Box 9.5 for continuous responses. However, an analogue to studentized residuals does not exist. In the case of generalized models, see the books of [McCullagh and Nelder \(1989\)](#), [Tutz \(2011\)](#), and [Collett \(1991\)](#) for the definition of appropriate residuals. In principle, influence measures and outlier analysis, as

well as Cook's distance, are also available. However, to date, they have not been thoroughly examined and are not part of common software packages.

There is a contemporary approach for automatic variable selection and model choice in generalized STAR models using goodness-of-fit criteria such as AIC or cross validation, as developed in [Belitz and Lang \(2008\)](#). The proposed algorithms are able to:

- Determine whether or not a particular covariate is to be included in the model
- Determine whether a continuous covariate enters the model linearly or nonlinearly
- Determine whether or not a spatial effect enters the model
- Determine whether or not a unit- or cluster-specific heterogeneity effect is included in the model
- select complex interaction effects (two-dimensional surfaces, varying coefficient terms)
- Select the degree of smoothness of nonlinear covariate, spatial, or cluster-specific heterogeneity effects

Particular emphasis is devoted to modeling and the selection of interaction terms. The approach is included in the software package BayesX (`stepwisereg` objects); also see the R interface `R2BayesX` described in [Umlauf, Kneib, Lang, and Zeileis \(2012\)](#). For automatic estimation and model selection in STAR models, also see Sect. 9.7 on boosting.

9.6.2 Inference Based on Mixed Model Representation

A second approach for estimating STAR models is based on their representation as mixed models. In Sect. 8.1.9, we already discussed the close relationship between penalization approaches and mixed models. We now briefly discuss the generalization to STAR models.

For every model component $f_j = V_j \gamma_j$ the vector γ_j can be decomposed as

$$\gamma_j = \tilde{X}_j \beta_j + \tilde{U}_j \tilde{\gamma}_j.$$

Through appropriate choices of the design matrices $\tilde{X}$ and $\tilde{U}$, we obtain a vector β_j of fixed effects and a vector $\tilde{\gamma}_j \sim N(\mathbf{0}, \tau_j^2 \mathbf{I})$ of independent and identically distributed random effects. Analogously, the vector of function evaluations can be expressed as

$$f_j = V_j(\tilde{X}_j \beta_j + \tilde{U}_j \tilde{\gamma}_j) = X_j \beta_j + U_j \tilde{\gamma}_j,$$

with $X_j = V_j \tilde{X}_j$ and $U_j = V_j \tilde{U}_j$. Applying this decomposition to all model components, we obtain a large mixed model whose fixed effects consist of the original parametric effects β and the fixed effects β_j , $j = 1, \dots, q$, resulting from the reparameterization. The random effects are composed of the vectors $\tilde{\gamma}_j$, $j = 1, \dots, q$, with joint distribution $N(\mathbf{0}, \text{diag}(\tau_1^2 \mathbf{I}, \dots, \tau_q^2 \mathbf{I}))$. Thus, the resulting mixed model is a variance components model. To estimate the model, in particular the variance or smoothing parameters, we can use the methodology

presented in Sect. 8.1.9. Note that the reparameterization generally causes redundant columns in the design matrix of the fixed effects. The reason for the redundancy is the identification problem of additive models. Every nonparametric model term has its own level corresponding to a constant in the fixed effects part of the reparameterization. After the reparameterization, the model then has these further constants, in addition to the intercept in β , so that the joint design matrix is no longer of full rank. In order to obtain an identifiable model, redundant columns need to be eliminated from the design matrix. For the constants, this mainly corresponds to the previously discussed centering conditions on f_j .

For additive and geoadditive models, inference based on mixed models is extensively discussed in [Ruppert, Wand, and Carroll \(2003\)](#). Mixed model inference in generalized STAR models is proposed in [Fahrmeir, Kneib, and Lang \(2004\)](#). Corresponding extensions to categorical responses are given in [Kneib and Fahrmeir \(2006\)](#). A detailed presentation can also be found in [Kneib \(2005\)](#).

Model Choice and Diagnosis

The remarks mentioned above regarding model choice and diagnosis mostly remain valid with inference based on mixed models.

To verify whether or not the flexible nonparametric modeling of the individual covariates effects is necessary, we can in principle apply statistical tests regarding the variance parameters. If we want to test, e.g., a P-spline with a second-order difference penalty against a linear effect, then this corresponds to the hypothesis

$$H_0 : \tau^2 = 0 \quad \text{versus} \quad H_1 : \tau^2 > 0.$$

At first thought, the likelihood ratio statistic that compares the likelihoods under H_0 and H_1 may be used to perform the test. However, the parameter to be tested is on the boundary of the parameter space under the null hypotheses. This contradicts the usual regularization conditions which are used when deriving the asymptotic χ^2 -distribution of the likelihood ratio test statistic. Thus, the likelihood ratio test statistic does not follow a χ^2 -distribution in this case. Currently, the exact distribution of the likelihood ratio statistic can only be derived in relatively simple special cases with Gaussian responses. Further information can be found, for example, in [Crainiceanu and Ruppert \(2004\)](#) or [Crainiceanu et al. \(2005\)](#). A simulation-based approach for obtaining p-values that works also for models with more than one variance parameter has been proposed in [Scheipl et al. \(2008\)](#) and is implemented in the R package `RLRsim`; see also Sect. 7.3.4, p. 381.

Similar problems with violated regularity conditions occur when considering the AIC for comparing models with estimated smoothing variance and the simpler model obtained when setting the variance equal to zero. This includes the important case of comparing a simple linear model for a specific covariate with a model comprising a P-spline with second-order difference penalty with smoothing variance estimated from the mixed model representation. More specifically, [Greven and Kneib \(2010\)](#) considered two variants of the AIC derived from the marginal and the conditional perspective on mixed models (compare Sect. 7.1.4) and investigated

their properties in this situation. The marginal AIC is obtained when integrating out the random effects and relying on the marginal normal distribution with correlations between the responses for deriving the likelihood involved in the AIC. It turns out that basic regularity conditions of the AIC are violated in this situation and therefore the marginal AIC cannot be used to compare models with estimated variance components and models where the variances are set to zero. Nevertheless, the marginal AIC is routinely returned by several mixed model implementations in statistical software such as the `nlme` package in R.

The AIC we have used so far in this book is the conditional AIC that is specified as

$$-2l(\cdot) + 2\text{tr}(\mathbf{H})$$

where $l(\cdot)$ denotes the log-likelihood of the estimated model (conditional on the random effects) and $\mathbf{H}$ is the corresponding hat matrix. While this standard definition is valid in case of given smoothing variances, it has a bias towards the more complex model in case of estimated smoothing variances. In fact, the conditional AIC always decides for the model with nonparametric effect when the smoothing variance is estimated to be non-zero. This defect results from the fact that uncertainty due to the estimation of smoothing variances is not taken into account in the conditional AIC. As a consequence, the conditional AIC in its standard form also does not enable proper model choice in situations where the decision problem corresponds to differentiating between estimated smoothing variances and smoothing variances set to zero. [Greven and Kneib \(2010\)](#) therefore introduce a corrected conditional AIC that circumvents this problem, but this corrected AIC is not yet available in standard software packages and is also restricted to models with Gaussian responses. We therefore give the following recommendations for practical work with the conditional AIC:

- If there is strong evidence for one model in the model comparison (i.e., if one model has significantly lower AIC), then the standard conditional AIC should be reliable enough since the problems described above mostly occur when the smoothing variance is close to zero.
- The standard conditional AIC can still be used to compare models that only differ in the specification for the parametric effect or that compare different structures for the random effect (e.g., an i.i.d. random effect versus a Markov random field for a spatial effect based on geographical regions).

In that sense, the model comparisons made so far in this chapter should be valid. The only borderline case appeared in Example 9.3. However, since we found a preference for the simpler model in this example anyway, the decision should still be fine.

9.6.3 Bayesian Inference Based on MCMC

Gaussian Response

We first assume Gaussian responses, i.e.,

$$\mathbf{y} \mid \boldsymbol{\gamma}_1, \dots, \boldsymbol{\gamma}_q, \boldsymbol{\beta}, \sigma^2 \sim N(\boldsymbol{\eta}^{struct}, \sigma^2 \mathbf{I}).$$

The density of the observation model is then proportional to the likelihood and is given by

$$p(\mathbf{y} | \boldsymbol{\gamma}_1, \dots, \boldsymbol{\gamma}_q, \boldsymbol{\beta}, \sigma^2) \propto \frac{1}{(\sigma^2)^{\frac{n}{2}}} \exp \left(-\frac{1}{2\sigma^2} (\mathbf{y} - \boldsymbol{\eta}^{struct})' (\mathbf{y} - \boldsymbol{\eta}^{struct}) \right).$$

Bayesian inference is based on the smoothness prior

$$p(\boldsymbol{\gamma}_j | \tau_j^2) \propto \left(\frac{1}{\tau_j^2} \right)^{\text{rk}(\mathbf{K}_j)/2} \exp \left(-\frac{1}{2\tau_j^2} \boldsymbol{\gamma}_j' \mathbf{K}_j \boldsymbol{\gamma}_j \right), \quad (9.17)$$

introduced in the Box given in Sect. 8.1.9 on p. 486 for the regression coefficients $\boldsymbol{\gamma}_j$ of the j th term of the structured additive predictor. For the variance parameters τ_j^2 , we specify an inverse gamma distribution with hyperparameters a_j and b_j , i.e.,

$$\tau_j^2 \sim \text{IG}(a_j, b_j),$$

as the prior distribution. For the variance σ^2 of the error term, we similarly define an inverse gamma distribution with hyperparameters a and b . In practice, small values are often chosen for a_j and b_j , e.g., $a_j = b_j = 0.001$ or $a_j = b_j = 0.0001$. This recommendation is only meaningful if the response values are in a specific range that is obtained by dividing the original response observations by their empirical standard deviation.

For the linear effects regression coefficients $\boldsymbol{\beta}$, in this chapter, we restrict ourselves to noninformative priors, i.e., $p(\boldsymbol{\beta}) \propto \text{const}$. A multivariate Gaussian prior $\boldsymbol{\beta} \sim \text{N}(\mathbf{m}, \mathbf{M})$ could alternatively be used, as seen in Sect. 5.6.

The density of the prior distribution of all parameters can now be written as the product of the densities in Eq. (9.17) and the densities of the inverse gamma distributions for the variance parameters. The posterior distribution is proportional to the product resulting from the likelihood with the prior distribution. We obtain

$$\begin{aligned} p(\boldsymbol{\theta} | \mathbf{y}) &\propto \frac{1}{(\sigma^2)^{\frac{n}{2}}} \exp \left(-\frac{1}{2\sigma^2} (\mathbf{y} - \boldsymbol{\eta}^{struct})' (\mathbf{y} - \boldsymbol{\eta}^{struct}) \right) \\ &\prod_{j=1}^q \frac{1}{(\tau_j^2)^{\text{rk}(\mathbf{K}_j)/2}} \exp \left(-\frac{1}{2\tau_j^2} \boldsymbol{\gamma}_j' \mathbf{K}_j \boldsymbol{\gamma}_j \right) \prod_{j=1}^q (\tau_j^2)^{-a_j-1} \exp \left(-\frac{b_j}{\tau_j^2} \right) \\ &(\sigma^2)^{-a-1} \exp \left(-\frac{b}{\sigma^2} \right), \end{aligned}$$

where $\boldsymbol{\theta}$ is the vector of all model parameters (including the variance parameters). We will next describe a Gibbs sampler (see Sect. B.5.3 in Appendix B) for drawing random numbers from the posterior. To do so, the entire parameter vector is decomposed into the blocks $\boldsymbol{\gamma}_1, \dots, \boldsymbol{\gamma}_q, \boldsymbol{\beta}, \tau_1^2, \dots, \tau_q^2, \sigma^2$. A requirement of the Gibbs-sampler is that the full conditionals of the blocks must represent known

distributions, from which sampling of random numbers is fast and easily available. Similar to and as already described in Sect. 8.1.9, the conditional densities of the regression parameters are multivariate normal distributions, and the conditional densities of the variance parameters are inverse gamma distributions.

More precisely, we have

$$\boldsymbol{y}_j | \cdot \sim N(\boldsymbol{m}_j, \boldsymbol{\Sigma}_j)$$

with expectation and covariance matrix

$$\begin{aligned} \boldsymbol{m}_j &= E(\boldsymbol{y}_j | \cdot) = \left(\frac{1}{\sigma^2} \boldsymbol{V}_j' \boldsymbol{V}_j + \frac{1}{\tau_j^2} \boldsymbol{K}_j \right)^{-1} \frac{1}{\sigma^2} \boldsymbol{V}_j' (\boldsymbol{y} - \boldsymbol{\eta}_{-j}^{struct}) \\ \boldsymbol{\Sigma}_j &= \text{Cov}(\boldsymbol{y}_j | \cdot) = \left(\frac{1}{\sigma^2} \boldsymbol{V}_j' \boldsymbol{V}_j + \frac{1}{\tau_j^2} \boldsymbol{K}_j \right)^{-1}. \end{aligned}$$

The vector $\boldsymbol{\eta}_{-j}^{struct} = \boldsymbol{\eta}^{struct} - \boldsymbol{V}_j \boldsymbol{y}_j$ is the current predictor $\boldsymbol{\eta}^{struct}$ without the j th term. Similarly, for the vector $\boldsymbol{\beta}$, we obtain a multivariate normal distribution with expectation and covariance matrix given by

$$\begin{aligned} \boldsymbol{m}_\beta &= E(\boldsymbol{\beta} | \cdot) = (\boldsymbol{X}' \boldsymbol{X})^{-1} \boldsymbol{X}' (\boldsymbol{y} - \tilde{\boldsymbol{\eta}}^{struct}) \\ \boldsymbol{\Sigma}_\beta &= \text{Cov}(\boldsymbol{\beta} | \cdot) = \frac{1}{\sigma^2} (\boldsymbol{X}' \boldsymbol{X})^{-1}, \end{aligned}$$

where $\tilde{\boldsymbol{\eta}}^{struct} = \boldsymbol{\eta}^{struct} - \boldsymbol{X} \boldsymbol{\beta}$ is the predictor $\boldsymbol{\eta}^{struct}$ without the linear effects. For the variance parameters, we obtain

$$\begin{aligned} \tau_j^2 | \cdot &\sim \text{IG}(a_j + 0.5 \text{rk}(\boldsymbol{K}_j), b_j + 0.5 \boldsymbol{y}_j' \boldsymbol{K}_j \boldsymbol{y}_j), \\ \sigma^2 | \cdot &\sim \text{IG}(a + 0.5n, b + 0.5(\boldsymbol{y} - \boldsymbol{\eta}^{struct})'(\boldsymbol{y} - \boldsymbol{\eta}^{struct})). \end{aligned}$$

The derivation of all full conditionals is similar to the classical linear model in Sect. 4.4. It is now straightforward to implement a Gibbs sampler that successively draws samples from the full conditionals. Numerically efficient sampling of random numbers from the high-dimensional Gaussian distributions is guaranteed with methodology for banded or sparse matrices. More details can be found in [Rue and Held \(2005\)](#), [Lang and Brezger \(2004\)](#), and [Lang et al. \(2012\)](#).

MCMC-Based Inference Using Data Augmentation

As already discussed in Sect. 5.6.3 (p. 316), the Gibbs sampler for Gaussian responses can be often modified for MCMC inference with non-Gaussian responses. Applying the data augmentation schemes of Sect. 5.6.3 to STAR models is straightforward; its description would be identical to Chap. 5 and is therefore omitted.

Non-Gaussian Responses

For some distributions, e.g., gamma-distributed responses, a Gibbs sampler can no longer be derived. In this case, MCMC inference can be based on MH algorithms with IWLS proposals as described for generalized linear models in Sect. 5.6. Details for generalized STAR models can be found in [Brezger and Lang \(2006\)](#) and [Lang et al. \(2012\)](#).

Model Choice and Diagnosis

For comparing competing models, we can use the DIC (see Sect. B.5.4 in Appendix B) as a global goodness-of-fit measure. The DIC can be computed as a by-product of the MCMC sampler. However, the DIC is subject to sampling variation and therefore shows (slightly) different values in each MCMC run. In the case that two competing models have very different DIC values, problems due to the sampling variability of the DIC should not be an issue. Some care has to be taken if DIC values are close to each other. In such a case, comparison is only possible by means of several (possibly time intensive) MCMC runs (at least 20) and then averaging the DIC values.

For model diagnosis, we can again use the various forms of residuals. With Bayesian inference based on MCMC methods, some additional diagnostic checks regarding convergence of the samplers are necessary. The literature provides a huge variety of diagnostic tools; refer to [Cowles and Carlin \(1996\)](#) and [Mengersen, Robert, and Guihenneuc-Jouyaux \(1999\)](#) for an overview. Their use is, however, restricted to relatively simple models with only a few parameters. In any case, including complex models with many parameters, it is extremely important to examine the sampling paths and autocorrelation functions of *all* samples. Striking features, especially non-convergence, low acceptance rates, and high autocorrelations, become apparent with these tools; see Example B.10 in Appendix B. When using the MCMC algorithms described in this chapter, problems usually only occur when the models that are being estimated are too complex for the data at hand.

Another important diagnostic step for Bayesian models is to perform a *sensitivity analysis* regarding the effect of slight changes in the priors (particularly the hyperparameters) on the posterior. In the case of STAR models, this requires examining the sensitivity of results on the choice of the hyperparameters a_j and b_j of the inverse gamma prior distribution for the variance parameters τ_j^2 . Depending on the choice of hyperparameters, it is possible that the estimated results differ considerably. This is especially the case if we do not have sufficient information, i.e., a limited number of observations in subdomains of the covariate range. In most cases, however, the results are relatively stable. The possible dependence on the choice of hyperparameters is not a disadvantage of Bayesian inference for STAR models. On the contrary, the Bayesian approach allows a careful sensitivity analysis when examining the stability of the chosen model.

Recently, a Bayesian approach based on spike and slab priors for STAR models, with built-in model and variable selection options, has been proposed; see [Scheipl](#),

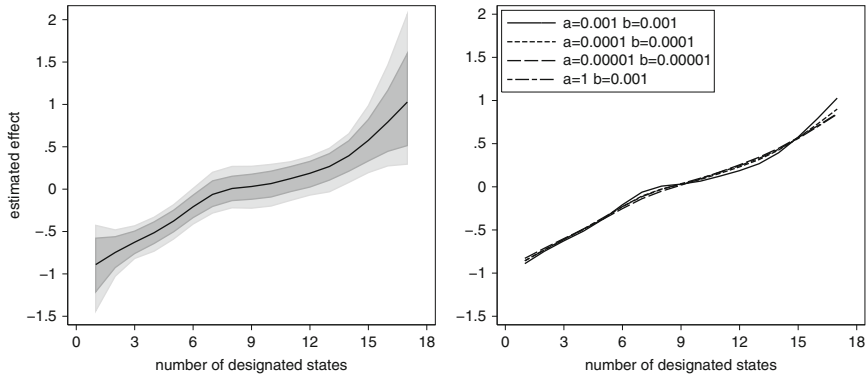


Fig. 9.14 Patent opposition: The *left panel* shows the estimated nonlinear effect of the number of designated states for the patent. The *right panel* shows the estimated effect of *ncountry* based on four different combinations of the hyperparameters *a* and *b* for the prior of the variance parameter

Fahrmeir, and Kneib (2012) for details. Software is provided in the R package `spikeSlabGAM`, outlined in Scheipl (2011).

Lastly, we want to mention that there are some extensions for the Bayesian adaptive procedures described in Sect. 8.1.10, which allow for simultaneous estimation and model choice. Since generally usable software for this methods is not available we restrain from presenting detailed algorithms but rather point to the references given in Sect. 8.1.10 for details.

Example 9.8 Patent Opposition—Additive Model

In the Examples 2.8 (p. 35), 5.1 (p. 275) and 5.4 (p. 288), we modeled the effect of the covariate *ncountry* on the binary response *opp* using a cubic polynomial. For comparison, we now estimate the effect $f(\text{ncountry})$ nonparametrically. This leads to the additive logit model

$$P(\text{opp}_i = 1) = \frac{\exp(\eta_i^{\text{add}})}{1 + \exp(\eta_i^{\text{add}})}$$

with the predictor

$$\begin{aligned} \eta_i^{\text{add}} = & f(\text{ncountry}_i) + \beta_0 + \beta_1 \text{yearc}_i + \beta_2 \text{ncit}_i + \beta_3 \text{nclaims}_i + \beta_4 \text{ustwin}_i \\ & + \beta_5 \text{patus}_i + \beta_6 \text{patgsgr}_i. \end{aligned}$$

The (possibly) nonlinear function $f(\text{ncountry})$ is modeled by cubic P-splines and estimated using MCMC. We included the centered year (variable *yearc*) rather than the original variable *year* because otherwise the intercept and the effect of year would be highly correlated. We used `bayesreg` objects of the software `BayesX`. Figure 9.14 (left) shows the posterior mean together with 80 % and 95 % (pointwise) confidence bands. Clearly, the function does not greatly differ from a straight line. We obtain $\text{DIC} = 2,993$. Modeling the effect of *ncountry* linearly or with a polynomial, we obtain $\text{DIC} = 2,993$ and $\text{DIC} = 2,995$, respectively. Thus, a linear effect is apparently sufficient. This also confirms the result of the statistical test conducted in Example 5.4.

The right panel of Fig. 9.14 shows the result of a sensitivity analysis, in which different hyperparameters a and b of the prior for the variance parameter τ^2 were tested. We used the combinations $(a = 0.001, b = 0.001)$, $(a = 0.0001, b = 0.0001)$, $(a = 0.00001, b = 0.00001)$, and $(a = 1, b = 0.001)$. We find that estimated results do not differ much from each other. Moreover, the figure once again confirms the linearity assumption for the effect of $ncountry$.

△

9.7 Boosting STAR Models

In Sect. 4.3, we introduced boosting as a regularized estimation technique that had the particular advantage to combine model estimation with automatic model choice and variable selection. Such a combination would be very desirable in the context of STAR models, especially since several competing modeling possibilities and model formulations are available. We will therefore extend boosting approaches to STAR models.

In the context of linear regression models, boosting estimates were obtained by repeatedly applying simple ordinary least squares fits to iteratively updated residuals. In *componentwise* boosting, only the best-fitting covariate was updated in each iteration of the boosting process. To generalize this componentwise approach to STAR models, we basically have to replace the least squares base-learners with penalized least squares base-learners.

Therefore, we assume a structured additive regression model with predictor

$$\eta_i = f_1(v_{i1}) + \dots + f_q(v_{iq}) + \beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik},$$

where as before the functions f_j may represent different types of nonlinear regression effects. In matrix notation, the model can be reexpressed as

$$\eta = V_1 \gamma_1 + \dots + V_q \gamma_q + X_1 \beta_1 + \dots + X_L \beta_L,$$

where we have grouped covariates with parametric effects in $L \leq k + 1$ suitable blocks of coefficients. This grouping may result, for example, from considering dummy or effect coding of categorical covariates where all dummy variables should be assigned to a common block, such that when performing model choice, either the complete block is selected or the complete block is deselected from the model. A similar situation arises when considering a polynomial expansion of a continuous covariate. In this case, it may be useful to combine all polynomials in one coefficient block. However, in the simplest situation, each block will only comprise exactly one covariate and therefore $L = k + 1$.

We further assume that the estimation problem at hand is described in terms of a suitable loss function ρ . Typically, this will be the least squares criterion in case of Gaussian responses or the negative log-likelihood for exponential family regression. More general situations are also conceivable, for example, to cast median or robust regression in the context of boosting.

Similar to the presentation in Sect. 4.3, the base-learning procedure for estimating parametric linear effects for the blocks $X_l\beta_l$ is given by ordinary least squares fits and can therefore be characterized by the corresponding hat matrices

$$H_l = X_l(X_l'X_l)^{-1}X_l', \quad l = 1, \dots, L.$$

Analogously, all base-learning procedures for nonlinear effects $V_j\gamma_j$ can be described in terms of penalized least squares fits as characterized by the hat matrices

$$H_j = V_j(V_j'V_j + \lambda_j K_j)^{-1}V_j', \quad j = 1, \dots, q,$$

with smoothing parameters $\lambda_j \geq 0$. Note that the smoothing parameters are now no longer part of the estimation process but will be specified a priori to assign a certain flexibility to the base-learners.

Based on these definitions, a generic boosting algorithm for structured additive regression models is given in Box 9.6.

In each iteration of the algorithm, all base-learning procedures are fitted to the current negative gradient vector $\mathbf{u}$, but only the best-fitting base-learner is actually updated. Note that this decision is always based on the L_2 loss (as indicated in Box 9.6) regardless of the specific loss function describing the estimation problem since the negative gradient vectors is treated as a continuous response in the base-learner. Updating only the best fitting base-learner allows us to implement model choice by setting an optimal, data-driven stopping iteration m_{stop} . Typically, the important and influential covariates will be selected first, while those covariates that are only weakly associated with the response will be selected only in very late boosting iterations. Hence, when stopping early enough, such covariates will effectively drop out of the estimated model. An optimal stopping iteration will usually be determined based on cross validation techniques.

Note that the model choice possibilities of boosting can be further improved when considering competing modeling alternatives for the same covariate. For example, a continuous covariate can be included linearly and nonlinearly in the candidate model such that in each iteration the algorithm compares the performance of the linear and the nonlinear modeling alternative and decides for the better fitting alternative. However, care has to be taken when actually implementing this approach since the nonlinear effect naturally contains the linear effect as a special case and therefore has a greater chance to be selected simply because the corresponding base-learner is more flexible. It is therefore important to make the base-learners comparable in terms of their complexity. This can be achieved by a suitable choice of the smoothing parameter in the penalized least squares base-learners. Note that these are no longer hyperparameters to be estimated but rather constitute the flexibility of the base-learner. Typically, user-specified degrees of freedom will be assigned to each base-learner and the corresponding smoothing parameter will be numerically determined accordingly. Assigning the same degrees of freedom to each base-learner yields a comparable complexity for all model terms. This may also require an assignment of ridge-type base-learners to the parametric

9.6 Boosting in STAR Models

- (i) Initialize all parameter blocks β_l and vectors of function evaluations f_j with suitable starting values $\hat{\beta}_l^{(0)}$ and $\hat{f}_j^{(0)}$ and set the predictor $\eta^{(0)} =$ offset with a suitable offset (such as the population minimizer of the loss function). Choose a maximum number of iterations m_{stop} and set the iteration index to $m = 1$.
- (ii) Compute the negative gradients (i.e., partial derivatives of the loss function)

$$u_i = - \frac{\partial}{\partial \eta} \rho(y_i, \eta) \Big|_{\eta = \hat{\eta}_i^{(m-1)}}, \quad i = 1, \dots, n,$$

that will serve as working responses for the base-learning procedures.

- (iii) Fit all base-learning procedures to the negative gradients to obtain estimates $\hat{b}_l^{(m)}$ and $\hat{g}_j^{(m)}$ and find the best-fitting base-learning procedure, i.e., the one that minimizes the L_2 loss

$$(u - \hat{u})'(u - \hat{u}),$$

inserting either $X_l \hat{b}_l^{(m)}$ or $\hat{g}_j^{(m)}$ for $\hat{u}$.

- (iv) If the best-fitting base-learner is the linear effect with index l^* , update the corresponding coefficient vector as

$$\hat{\beta}_{l^*}^{(m)} = \hat{\beta}_{l^*}^{(m-1)} + \nu \hat{b}_{l^*}^{(m)},$$

where $\nu \in (0, 1]$ is a given step size, and keep all other effects constant, i.e.,

$$\hat{\beta}_l^{(m)} = \hat{\beta}_l^{(m-1)}, \quad l \neq l^* \quad \text{and} \quad \hat{f}_j^{(m)} = \hat{f}_j^{(m-1)}, \quad j = 1, \dots, q.$$

Correspondingly, if the best-fitting base-learner is the nonlinear effect with index j^* , update the vector of function evaluations as

$$\hat{f}_{j^*}^{(m)} = \hat{f}_{j^*}^{(m-1)} + \nu \hat{g}_{j^*}^{(m)}$$

and keep all other effects constant, i.e.,

$$\hat{\beta}_l^{(m)} = \hat{\beta}_l^{(m-1)}, \quad l = 1, \dots, L, \quad \text{and} \quad \hat{f}_j^{(m)} = \hat{f}_j^{(m-1)}, \quad j \neq j^*.$$

- (v) Unless $m = m_{\text{stop}}$, increase m by one and go back to (ii).

effects of categorical covariates with a large number of levels to prevent a preference for coefficient blocks having a larger number of coefficients and therefore more flexibility.

A further improvement of the model choice capabilities of boosting can be achieved when decomposing certain types of effects. For example, a nonlinear effect $f(z)$ of a continuous covariate z may be represented as

$$z\beta + \tilde{f}(z),$$

i.e., as the sum of its linear effect $z\beta$ and the deviation from the linear effect $\tilde{f}(z) = f(z) - \beta z$. This has the advantage that the boosting algorithm can automatically differentiate between no effect of z (neither the linear nor the nonlinear effect are selected), linear effect of z (only the linear part is selected), and nonlinear effect of z (the nonlinear part is also selected). Technically, such a decomposition can be accomplished based on the mixed model representation of penalized least squares estimates introduced in Sect. 8.1.9.

We will illustrate these extended model choice abilities of boosting, as well as the general principles of the algorithm for structured additive regression models in the following case study on malnutrition in Zambia.

9.8 Case Study: Malnutrition in Zambia



Using the data on malnutrition in Zambia this section illustrates how the methods presented in this book can be applied in practice. In the next section, we first provide some general guidelines on how to carry out a regression analysis. We then present the case study on malnutrition in Zambia.

9.8.1 General Guidelines

Regression problems can be generally divided into the following main steps:

Descriptive Analysis of Raw Data

The first important step in *every* statistical application is the careful analysis of the raw data. We first need to get an overview of the (univariate) distribution of all variables in the data set. Useful tools are the graphical devices and summary statistics outlined in Sect. 1.2.1. The following goals are pursued in this first analysis:

- *Description of the distribution of all variables in the data set.*
- *Discovery of data anomalies.* This includes incorrect coding, implausible and extreme data values. In regression problems, the inspection of the covariate domain is especially important. For some continuous covariates the observed data are often sparse, particularly near the boundaries. With such limited information in some regions, it is often difficult to make valid statements regarding the effect

of a covariate on the response. Sometimes it is even appropriate to exclude extreme observations from further analysis. A typical example was the analysis of the data on patent opposition in Example 2.8 (p. 35), where we could observe an extremely positively skewed distribution for the covariate *nclaims*. The majority of observations lie in the region 0–60. However, some observations were much larger, with the maximum value of 355. Due to the fact that very little information is available in the interval 61–355, all observations with *ncountry* > 60 were excluded from further analysis.

The problem of unequally distributed values can also exist with categorical covariates. For some categories, absolute or relative frequencies are often very low. In the extreme case, a covariate may be useless for regression analysis. This is, for example, the case with binary covariates when one of the two categories is not populated at all, or perhaps only very sparsely. The missing or limited variability of the covariate then precludes examination of the effect on the response variable. In many cases, the problem of sparsely populated categories can be solved by combining individual categories (when meaningful).

Data Preparation

After the initial analysis of the raw data, the data must be prepared for the regression analysis. This means that some extreme observations, possibly even some covariates, may be excluded from further analysis. Outlier analysis and plausibility checks are useful tools to assist in this regard. Furthermore, existing data errors need to be corrected, and possibly new covariates derived from existing covariates need to be created (e.g., orthogonal polynomials or other transformations).

Finally, we must ensure that the raw data does not get lost during data preparation. This may sound rather trivial, but lost raw data are a frequent phenomenon in practice. It is strongly recommended to create a new data set out of the raw data containing the prepared observations and variables ready for regression analysis. Also, the necessary steps for data preparation should be recorded and documented.

Graphical Two-Dimensional Correlation Analysis

In the case of continuous responses, two-dimensional graphical correlation analyses between the response values and each covariate can provide important information about the type and strength of the relationship. However, any interpretation should be taken with care, since at this stage we are only investigating two-way correlations, and we are generally not controlling for the effect of other covariates. Hence, it is absolutely possible that the further regression analysis provides different results. Useful devices for the correlation analysis are the graphical tools outlined in Sect. 1.2.2 (scatter plots, box plots). We should also take care to find possible outliers (see Sect. 3.4.4). Often—but not always—such observations can be identified with the help of scatter plots.

In the case of discrete responses, in particular categorical variables, graphical exploratory tools can be of rather limited use.

Estimation of Preliminary Working Models

The semiparametric regression models discussed above and in the previous chapter are particularly useful for exploratory data analysis and for getting a first working model. With these models, nonlinear covariate effects, as well as spatial and cluster-specific heterogeneity, can be detected quickly and automatically. Usually, the starting point is a pure main effects model. Interactions can be considered in subsequent steps. In addition to the results from the univariate and bivariate exploratory analyses, expertise knowledge and knowledge from former examinations should be considered in the model building process. It is generally not advisable to include all available variables in an unreflected way into the regression model.

Generally, a number of competing model specifications will be available. For example, it is possible to have models with and without spatial components or perhaps the effect of a variable is doubtful, so we may estimate models with and without the covariate of concern. We can then compare the competing models with a goodness-of-fit or model choice criterion. Based on this criterion, model specifications with little impact on the fit can then be excluded from further analysis.

In the end, there is generally not the *one perfect model* which dominates the other models in all aspects. Rather there are several competing models which fit the data (almost) equally well.

Model Diagnostics, Model Evaluation, and Improvement of the Working Models

The main goal of this step is to improve or to refine the models of the previous step. The following steps can be taken:

Reduce to a Simple Parametric Model

In a number of cases, the covariate effects, which were discovered through the semiparametric analysis, can be nearly linear or rather simple nonlinear functions. In such a case, parts of the model can be replaced by simple parametric terms, even using piecewise linear or constant terms.

Exploratory Analysis of the Spatial or Cluster-Specific Effects

The spatial or cluster effects can be interpreted as surrogates for unobserved or unavailable covariate effects. In some cases, a careful analysis of the heterogeneity effects provides information about new covariates or aspects, which thus far had not been taken into consideration. For example, when analyzing the claim frequency of the vehicle insurance data (see Example 2.12, p. 52), we discovered that the claim frequency is higher in urban than in rural areas. In such a case, we might be able to replace the nonparametric spatial effect with a binary dummy variable, which identifies the urban areas. The ideal outcome is, in any case, the replacement or at least the decrease of the spatial or cluster-specific effects. This can be achieved by including new and reasonably interpretable covariates derived by investigating the heterogeneity effects.

As Example 9.7 (p. 556) shows, the inclusion of spatial or cluster-specific effects can lead to identifiability problems. In particular, the effects of other covariates can be concealed by the spatial or cluster effects. Hence, it is advisable to carefully examine the differences in results of models with and without spatial or cluster-specific components. As mentioned, this problem is subject to growing research; see the references given on p. 556.

Model Diagnostics

Model diagnostic tools aim at checking the adequacy of a working model and whether certain assumptions, such as homoscedasticity of the error terms, are fulfilled. This may ultimately lead to the consideration of alternative model formulations and appropriate refinements of the working model. For parametric linear models, there are many useful model diagnostic tools; see Sect. 3.4.4. For semiparametric models, many of the tools, in particular residual analysis, can be easily adapted. Some diagnostics for outlier analysis are, however, either not at all or methodologically less developed or not included in software packages. In models with discrete responses, the use of residual plots is also limited.

In the case of continuous responses, residual plots provide information about omitted or misspecified covariate effects, outliers, and heteroscedastic or correlated errors. Depending on the findings, different alternatives are conceivable:

- *Misspecified covariate effect.* In case of misspecified covariate effects, we have the following options:
 - *Nonlinear effect.* The effect of a continuous covariate, initially modeled as linear, proves to be nonlinear. In this case, more flexible modeling of the effect, e.g., using spline-based methods is in order.
 - *Missing covariates.* Available covariates that were initially not incorporated can be included in the model at any time. The situation is more difficult with unobserved covariates. In this case, spatial or cluster-specific effects can be used for modeling unobserved heterogeneity.
 - *Missing interactions.* Complex interactions can be estimated either with the parametric approaches discussed in Sect. 3.4, or with the nonparametric alternatives discussed in Sect. 9.3. However, interactions should not be unnecessarily used, as they can greatly increase the complexity of the model.
- *Heteroscedastic errors.* In the presence of heteroscedastic errors, we can generally use the alternatives discussed in Sect. 4.1.3 (p. 186), i.e., variance stabilizing transformation of the response, two-step estimation, or simultaneous estimation using, e.g., the R package `gamlss`. The latter is restricted to purely additive models. In parametric linear models, we can also use the corrected covariance matrix of White presented in Sect. 4.1.3 on p. 190.
- *Correlated errors.* As discussed in Sects. 3.1.2 and 4.1.4, the existence of autocorrelation in the residuals indicates model misspecification. The primary goal should be to eliminate the misspecification as much as possible. The use of semiparametric models with correlated error terms is still in the development phase. For instance, identifiability problems are to be expected, e.g., when a

nonparametric time trend and correlated errors are simultaneously modeled; see [Fahrmeir and Kneib \(2008\)](#).

- *Outliers*. In case of outliers, we refer to the guidelines discussed in the context of linear models, see Sect. 3.4.4, p. 163. However, the use of robust procedures is limited with semiparametric models as the methodology is much less developed for this model class. A notable exception is median regression as a special case of quantile regression which in turn has undergone rapid development in recent years. Apart from the search for input data errors and/or possible explanations, probably the most important measure in the context of outliers is to refit the model without the outlying observations and to describe the differences in model fit.

Check Stability of Estimates Under Different Estimation Concepts

In many cases the models under consideration can be estimated using alternative estimation concepts. For instance, variable selection in the linear model can be performed by minimizing a model choice criterion (Sect. 3.4), with the LASSO (Sect. 4.2.3), using boosting (Sect. 4.3) or based on Bayesian model averaging (Sect. 4.4). (Generalized) linear mixed models can be estimated using maximum likelihood (Sect. 7.3), or fully Bayesian inference (Sect. 7.4). In structured additive regression models estimation may be based on either penalized likelihood in combination with the optimization of a model choice criterion (Sect. 9.6.1), a mixed model representation (Sect. 9.6.2), a fully Bayesian approach (Sect. 9.6.3), or boosting (Sect. 9.7). We generally recommend to estimate the models under consideration with as many estimation concepts as possible to check the stability of estimation results.

Presentation of Results

As already mentioned, the result of a regression analysis is generally not the one “best” model which dominates the other models in all aspects. Rather there are a few models which fit the data (almost) equally well. Usually, these models only differ in some minor aspects, which often are not relevant to the problem. When outlining and describing the results, the common core of all models should be pointed out. On the other hand, the differences between the competing models should also be presented. Regarding these differences, we face uncertainty about our final conclusions.

Finally we want to point out that any analysis should be comprehensible and reproducible for those who did not participate in the investigation. It is also essential that the software, in particular the code implemented for data analysis, is archived and well documented. Compare also [Koenker and Zeileis \(2009\)](#) who propose software tools that may be helpful for reproducible research.

9.8.2 Descriptive Analysis

When presenting the malnutrition data in Chap. 1, we already provided summary statistics and graphical visualization for the variables in the data set; see Table 1.3 (p. 7) and Fig. 1.4 (p. 14). The results can be interpreted as follows:

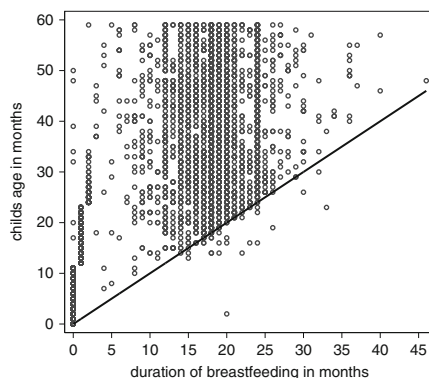


Fig. 9.15 Malnutrition in Zambia: *scatter plot* between the age of the child and the duration of breast-feeding. Observations below the main diagonal are implausible

- *Z-score*: The Z-score has a nearly symmetric distribution with values between -600 and 503 , with the majority of the observations between -600 and 300 . The average Z-score is -171.19 , i.e., on average the children in Zambia are much smaller compared to children of the same age in the reference population. Implausible or extreme values cannot be found.
- *Child's age and duration of breast-feeding*: Since the duration of breast-feeding cannot last longer than the age of the child, these two variables are related and should be considered together.

A preliminary univariate analysis reveals that age is nearly uniformly distributed. There are only slightly fewer older children than younger ones in the data set. The unusually rough form of the histogram appears to be more of an artifact, as is confirmed by a closer look at the frequency table or through refitted histograms based on different bandwidths.

Duration of breast-feeding varies in the data set from 0 to 46 months. The distribution is clearly bimodal with modes at 0 and 18 months. Almost 25 % of the children were never breastfed. It is very rare that the duration of breast-feeding lasts longer than 30 months (only 24 observations).

As a plausibility check, Fig. 9.15 includes a scatter plot between the age of the child and the duration of breast-feeding. As the duration of breast-feeding cannot last longer than the child's age, all observations below the bisecting line are implausible and thus are excluded from further analysis. The observations with $c.breastf > 30$ are also excluded for the following two reasons: First, breast-feeding rarely lasts longer than 30 months; secondly, two-dimensional smoothers are used for modeling the effect of age and duration of breast-feeding. Plausible results with two-dimensional smoothers are obtained only with a solid data base.

- *Mother's age at birth*: The mother's age varies from 13 to 48 years, with a mean value of 26.40 years. The distribution is clearly skewed to the right. Some very small and large values are unusual, but not impossible.

- *Mother's height*: The mother's height varies between 134 and 185 cm, with a mean value of 158.06 cm. The distribution is almost symmetric, without any apparent data anomalies.
- *Mother's BMI* : The body mass index is a known (but somewhat controversial) measure for the nutritional status of adults. A BMI between 18.5 and 25 is considered to be normal. A BMI less than 18.5 is an indication of underweight. A BMI between 25 and 30 is an indication of (some) overweight. A BMI of 30 is an indication of extreme overweight or obesity. In our data set, the BMI values vary from 13.15 to 39.29, with a mean value of 21.99. The majority of mothers are in the range of normal weight to minor overweight. A percentage that should not be neglected is that of underweighted mothers (nearly 10%). A smaller percentage of mothers are extremely overweight (about 2.6%). No particularly implausible or striking values are found in these data.
- *Mother's education and work status*: The predominant majority of the mothers (about 80%) have not completed their education. About 17% have completed primary education, and only 1.75% have a higher school degree. Slightly more than half of the mothers work, about 45% of the mothers do not work. In both variables, all categories have sufficient data, so that it is neither necessary to combine, nor exclude, categories.

In Chap. 1, we also performed graphical correlation analyses, as seen in Figs. 1.8 (p. 17) and 1.9 (p. 18). The plots can be interpreted as follows:

- *Child's age*: Initially the average Z-score decreases linearly and then remains almost constant for children of age higher than about 15–20 months. The observed relationship is postulated similarly in the malnutrition literature. It is assumed that the nutritional status is best during the first 6 months after birth since the child is well nourished first in the womb, and then after birth through the mother's milk. From this point a deterioration in the nutritional status is expected, as the child is more and more exposed to insufficient, qualitatively inferior, and more unsanitary food. Some stabilization is expected during the child's second year, but on a very low level.
- *Duration of breast-feeding*: The WHO recommends to breast-feed newborns for at least 12 months, since the mother's milk is the best form of nutrition for the baby during the first year. The literature about malnutrition, thus, assumes a generally positive effect of breast-feeding. If children are breast-fed longer than for 1 year, it is assumed that the effect decreases or may even become negative. Breast-feeding for a long time is then understood as an indicator for insufficient food supply.

In the scatter plot a relation between the Z-score and the duration of breast-feeding is barely detectable. There is a slight tendency towards a linearly decreasing relationship, i.e., with an increase in duration of breast-feeding, the average Z-score decreases. This would, however, contradict the literature. In any case, we should be careful when evaluating the effect of the duration of breast-feeding as we have to expect an interaction with the child's age. The effect of 1 month of breast-feeding for a 1-month-old child should differ from the same effect on a child of 3 years. In the first case, the duration of breast-feeding for

1 month is absolutely normal. In the second case, breast-feeding was stopped very early. We will therefore model the effect of child's age and duration of breast-feeding with a two-dimensional surface.

- *Mother's BMI*: The literature suggests an inverse U-form for the effect of the nutritional status of the mother, i.e., children of mothers with underweight or extreme overweight show a worse nutritional status than children of mothers who are of normal weight. The scatter plot shows largely a linear relationship with the Z-score, i.e., on average the Z-score increases linearly with BMI. From a BMI of 30–35 upward, i.e., in the case of obese mothers, the average effect seems to decrease, so that the postulated inverse U-form is reflected to some extent.
- *Mother's age at birth*: It seems as if the age of the mother has a weakly linear effect, i.e., the older the mother at the time of birth, the better the average nutritional status. In fact, the literature assumes that older mothers may take better care of a child than younger ones. These women have a greater chance of having completed vocational education, and they have in general more work experience leading to a higher income.
- *Mother's height*: It is well known that the height of the parents has an effect on the height of the children. Since the age-standardized height is used as a measure for the nutritional status, it is not surprising that we find a linearly increasing effect for the height of the mother in the scatter plot.
- *Education and work status*: We find a better nutritional status for children of better educated mothers. A relationship with the employment status cannot be observed.
- *Sex*: Gender-specific differences are not detectable (plot is not provided).

9.8.3 Modeling Variants

Based on the formulated hypotheses for the effect of the explanatory variables and the results of the descriptive analysis, the following modeling variants were examined. In order to be able to present the results in a compact way, we differentiate between different aspects of the models using the corresponding symbols:

- *Spatial effect*: Possible spatial heterogeneity was modeled either by a Markov random field (modeling variant M), by i.i.d. random effects (R), or parametrically using dummy variables for variable *region* (P).
- *Child's age and duration of breast-feeding*: As stated, the effect of the duration of breastfeeding interacts with the child's age. Therefore, the effect of both variables was estimated by a two-dimensional surface based on kriging (modeling variant I). We preferred kriging over other two-dimensional smoothers, for example, two-dimensional P-splines, as kriging appears to be most appropriate for the non-rectangular range of values associated with these two variables; see Fig. 9.15. Besides the two-dimensional modeling approach, also a pure nonparametric effect of age excluding the effect of the duration of breast-feeding was considered (modeling variant N). We used a P-spline with 20 interior knots based on a second-order difference penalty.

- *Mother's age at birth, height, and BMI*: The effects of the remaining continuous variables were modeled in four different ways:
 - All effects in a nonparametric way with P-splines (model variant N)
 - Nonparametric effect for the mother's body mass index and linear effects for the two other variables (B)
 - Assuming a quadratic polynomial for the body mass index and a linear effect for the two other variables (Q)
 - Assuming all effects to be of simple linear, parametric form (P)
- *Categorical covariates*: Every model includes the categorical variables *c_gender*, *m_education*, and *m_work* using appropriate dummy variables

The examined models can be uniquely characterized by a combination of three letters. For instance, the combination (MIN) denotes the model with a Markov random field for the spatial effect, a kriging approach for the two-dimensional interaction effect of age with the duration of breast-feeding, and nonparametrically modeled effects of the remaining continuous covariates. The model (RNQ) consists of i.i.d. random effects for the spatial effect, a nonparametric effect of the child's age, a quadratic effect for the mother's body mass index, and a linear effect for the remaining covariates.

9.8.4 Estimation Results and Model Evaluation

All models under consideration were estimated using mixed model technology and with `remlreg` objects of the software `BayesX`. Refer to Sects. 8.1.9 and 9.6.2. Also, we use the (conditional) AIC to assess the goodness of fit. For comparison, the models were also estimated with a fully Bayesian approach using MCMC techniques and `bayesreg` objects of the software `BayesX`. The differences in the resulting estimates are relatively small, so that the following presentation of results is restricted mostly to mixed model based inference. We will point out cases with noteworthy differences.

Of all examined models, the model (MIB), based on a Markov random field for the spatial effect, two-dimensional kriging for the effect of the child's age and the duration of breast-feeding, a nonlinear BMI effect, and otherwise linear effects, has minimal AIC. For all models, Fig. 9.16 shows AIC differences relative to the AIC of this "best" model. The models are sorted in ascending order according to their AIC values, so that models performing better in terms of AIC appear on the left. For the sake of comparison, a strictly parametric model was fitted as well. In this model, the age effect was modeled through a linear spline with one interior knot at $k_{age} = 23.5$. For this model, we obtain an AIC difference of 308, which is by far the worst fitting model.

Figure 9.16 reveals the following conclusions:

- The models which include a two-dimensional kriging effect of age and the duration of breast-feeding are superior to the models with only a nonparametric effect of the child's age. The differences in AIC are large between these two types of models so that we can safely decide for the model comprising an interaction term.

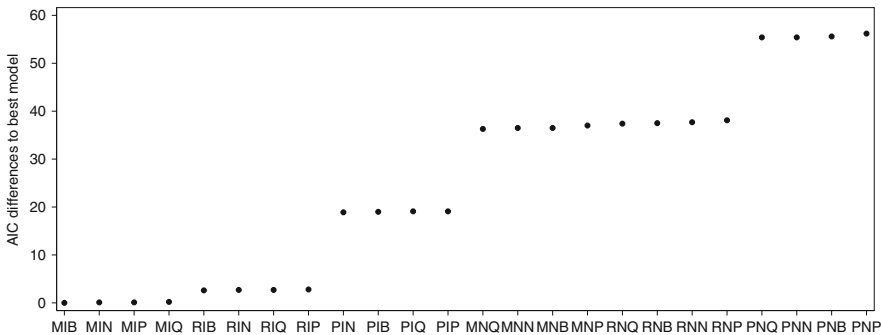


Fig. 9.16 Malnutrition in Zambia: AIC-differences to the best model for various model specifications

- Regarding the spatial effect, we obtain the best results when using a Markov random field. The use of i.i.d. random effects provides slightly higher AIC values of approximately 2.5 points. The parametric approach based on region dummies is not competitive. Therefore we can conclude that a more detailed spatial effect than the one based on dummy variables is indeed needed, but that the correlation structure specified for the spatial effect does not make too much of a difference.
- As Fig. 2.15 (p. 51) shows, the effects of the remaining three continuous variables are very smooth and do not differ much from a straight line. Therefore the four modeling variants for these variables have approximately the same AIC values. In all cases, the estimated effects are more or less indistinguishable. Only for the mother’s BMI there is uncertainty whether a linear or a weakly nonlinear effect is more appropriate. In this situation, the restrictions of the conditional AIC in Sect. 9.6.2 (p. 567) have to be taken into account when interpreting the results. There is a bias in the conditional AIC towards more complex models and therefore we would probably decide for linear effects in case of age and height of the mother due to the very small difference in the AICs between linear and nonlinear models. For the effect of the BMI of the mother, the situation is less clear and the detected minor nonlinearity should only be interpreted with care. It can then be useful to consider alternative estimation approaches or model choice techniques to validate the results from the AIC (and we will do so in the following section).

Figures 9.17 and 9.18 show the estimated effects of the continuous covariates and the spatial effect for the model (MIN), where the effects of all continuous variables were modeled in a nonparametric way. This is the model with the second best AIC value. Table 9.4 shows the results for the categorical covariates. We finally point out that Fig. 2.15 (p. 51) illustrates the results for the model (MNN), which differs from model (MIN) in that it does not consider the duration of breast-feeding.

- The results can be interpreted as follows:
- *Child’s age and duration of breast-feeding:* Initially the average Z-score decreases linearly as age increases. For children older than 18 months, the

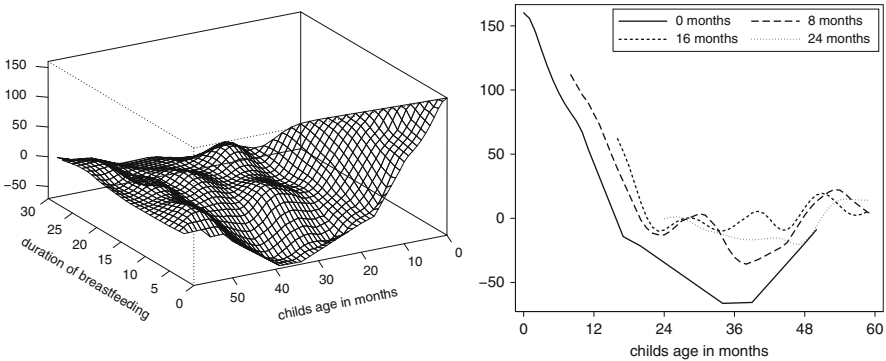


Fig. 9.17 Malnutrition in Zambia: interaction of the child's age and duration of breast-feeding in the model (MIN) (*left*). The *right* panel shows the age effect for various durations of breast-feeding

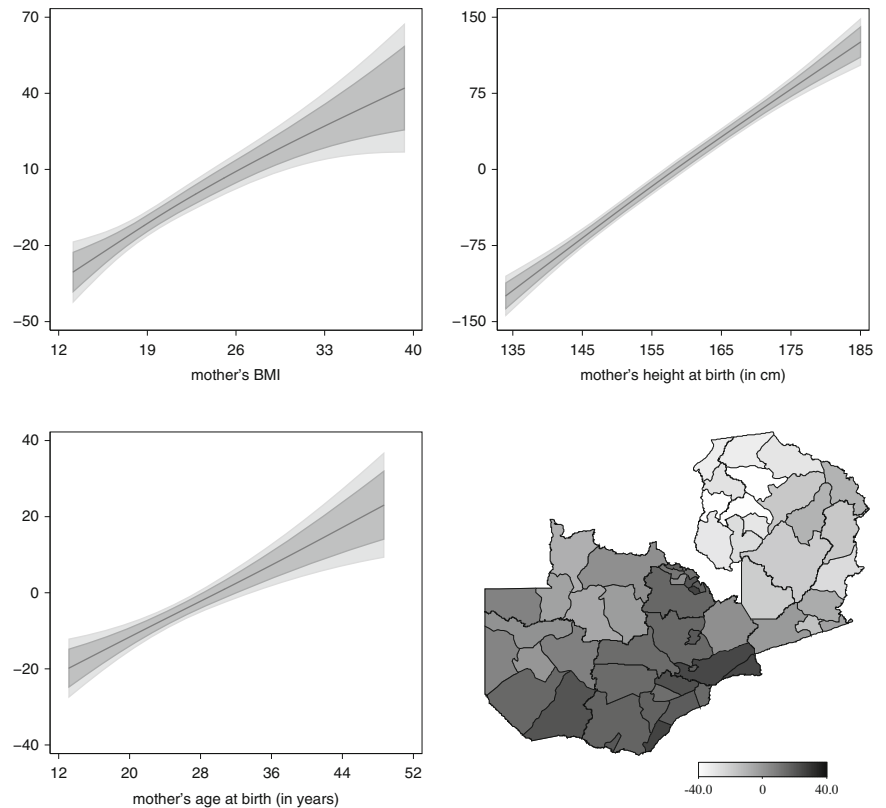


Fig. 9.18 Malnutrition in Zambia: nonparametric effects and spatial effect for the model (MIN)

Table 9.4 Estimated parametric effects in the model (MIN)

Variable	Coefficient	Standard-error	95 % confidence-interval	
<i>intercept</i>	−186.926	22.685	−231.398	−142.454
<i>c_gender</i>	−15.144	3.606	−22.213	−8.075
<i>m_education1</i>	−11.043	5.042	−20.927	−1.158
<i>m_education3</i>	21.774	5.055	11.863	31.684
<i>m_education4</i>	80.814	14.184	53.008	108.620
<i>m_work</i>	−4.420	3.807	−11.883	3.044

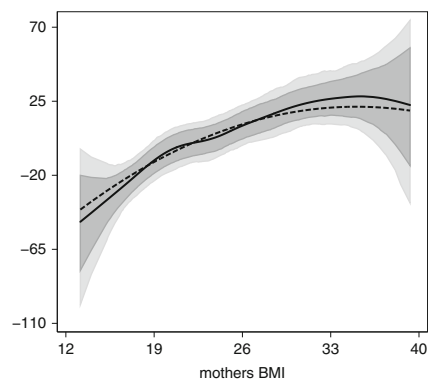


Fig. 9.19 Malnutrition in Zambia: estimated effect of the mother’s BMI for some alternative model specifications. The graph displays a nonlinear estimator (including 80 % and 95 % credible intervals) based on a fully Bayesian approach estimated with MCMC techniques. Additionally included is a quadratic polynomial (*dashed line*) in the model (MIB)

- nutritional status stabilizes at a low level. For children older than 3 years, a slight improvement can be observed. The curve for children who were never breast-fed is clearly below that of children who were breast-fed for 8, 16, and 24 months. We, thus, observe a positive effect of breast-feeding. However, differences regarding the duration of breast-feeding are not visible. Overall, the hypotheses formulated above can be mostly confirmed.
- *Mother’s BMI*: The estimated effect demonstrates that the nutritional status of children with underweight mothers is worse when compared to children of mothers who are of normal weight or mothers who are slightly overweight. For children of obese mothers, the strong linear effect diminishes. Based on these results, the average nutritional status is not decreasing with obese mothers as was initially postulated with the inverse U-form of the effect.

However, within the range of obese women, we find much wider confidence intervals caused by a small sample size. In this situation, it can be useful to take a look at the results of alternative models. In the model (MIQ), with AIC value almost equal to the best model, the BMI effect is modeled by a quadratic polynomial. Here, a tendency towards the postulated inverse U-form is visible

and the average Z-score slightly decreases for extremely overweight women, as provided in Fig. 9.19. For the effect of mother's BMI, we also find the most visible differences compared to fully Bayesian estimation based on MCMC methods (Fig. 9.19): We obtain an even more pronounced U-form compared to the quadratic polynomial.

What conclusions can be drawn from these contradictory results? It is common practice to present the results that are closest to the initially formulated hypotheses (in our case the results from the Bayesian approach) and to dismiss all other results. This is of course not correct. We recommend to point out the common core of the various results and then to describe the differences among them. In our case study, we observe an almost linear effect for mothers with underweight, normal weight, and slight overweight. For obese mothers, the BMI effect is less clear. The linear effect diminishes or even reverses and slightly decreases. For an effect of inverse U-form effect postulated in the literature, we have some evidence but no compelling proof. To obtain more precise results, we could, for example, overrepresent extremely overweight mothers in subsequent studies. If this is not possible, we could further examine data from other countries. We will reiterate the question on the precise form of the effect of mother's BMI in the next section when discussing approaches for automatic function selection.

- *Mother's height:* This effect is clearly linear. The taller the mother, the taller the child is, on average, and so is the Z-score. This effect, however, does not provide any conclusions about the nutritional status. On the contrary, this relatively large effect, that is mostly onset through heredity, questions the Z-score as (the only) measure for chronic malnutrition.
- *Mother's age at birth:* This effect is essentially linear as well. On average, older mothers have better nourished children. For this variable, we also found slight differences in results for the alternative Bayesian approach, which finds a steep increase of the average Z-score for mothers older than about 40 years of age. Thus, the positive effect would increase for very old mothers. Hence, our results are consistent with the hypotheses formulated in the previous section.
- *Spatial effect:* We notice a pronounced north–south gradient, i.e., the nutritional situation is much better in the north relative to the south. One reason for these pronounced differences could be that the climatic conditions are worse in the south, as these regions show a lower altitude than those in the north.
- *Categorical covariates:* The estimated parameters associated with the education dummies demonstrate a strong education effect. The more educated the mother, the better is the nutritional status of the child. The difference in Z-score between children whose mothers have completed primary education, compared to those whose mothers have not completed education, is on average only 11 points. The differences for children, whose mothers have been further educated are much more pronounced. Children whose mothers have higher education show on average a Z-score that is 90 points higher compared to children whose mother has not completed primary education.

The employment status of the mother does not seem to be of great importance. The effect is clearly nonsignificant.

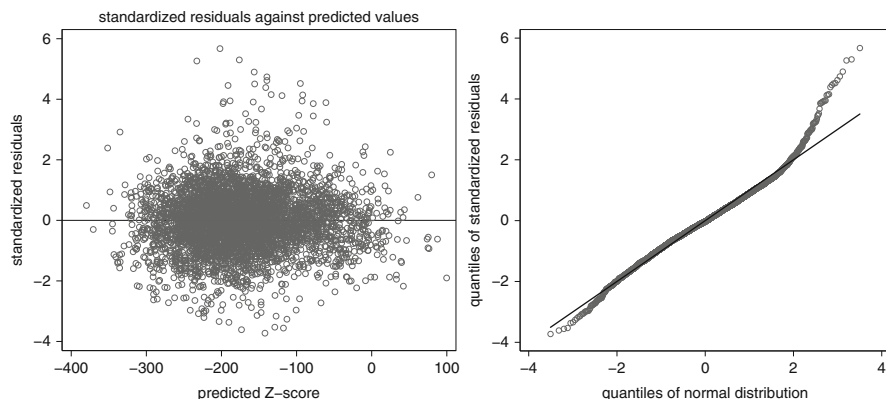


Fig. 9.20 Malnutrition in Zambia: standardized residuals versus predicted values (*left*) and a QQ-plot of standardized residuals (*right*) in the base model

As often stated in the literature, girls seem to be more robust than boys. Boys have a Z-score which on average is 15.14 points lower than girls.

Taking a look at the standardized residuals against the predicted Z-scores provides evidence about existing heteroscedasticity (left panel of Fig. 9.20). A two-step estimation approach that considers heteroscedasticity does not provide any noteworthy differences in results. The QQ-plot of the residuals provides clear evidence about departures from the normality assumption, especially for very large residuals with values greater than 2. This finding mainly concerns children with a positive Z-score, i.e., children with a very good nutritional status. Generally, minor violations of the normality assumption are not problematic since estimators are often relatively robust against such departures from normality. Possible alternatives for further analysis could be approaches that allow for simultaneous modeling of the mean and the skewness of the distributions; see our sketch of additive models for location, scale, and shape (GAMLSS) in Sect. 2.9. More details can be found in [Rigby and Stasinopoulos \(2005\)](#), [Rigby and Stasinopoulos \(2009\)](#), and the GAMLSS homepage <http://gamlss.org/>.

9.8.5 Automatic Function Selection

In the previous section, we have manually implemented function selection based on the comparison of different model specifications using AIC. However, recent developments in function selection also allow for the simultaneous determination of the unknown effects in a structured additive regression model combined with automatic function selection. In the following, we will illustrate such an approach relying on boosting utilizing function `gamboost` from the R package `mboost`. The approach proposed in [Belitz and Lang \(2008\)](#) yields comparable results.

For the boosting approach, we make use of the penalized least squares base-learners approach described in Sect. 9.7. Therefore, we define a model of maximum

complexity that comprises all candidate terms that should be considered in the component-wise fits during each boosting iterations. More specifically, we include the following effects:

- All categorical covariates (gender of the child, education, and work status of the mother) are represented as dummy-coded variables, where the complete block of dummies for each variable is treated as one term that can be selected or deselected in each boosting iteration.
- The interaction between age of the child and duration of breast-feeding is modeled in a kriging specification exactly as specified in the previous section.
- All remaining continuous covariates (mother's age at birth, height, and BMI) are modeled as potentially nonlinear effects based on penalized splines. To allow for a more detailed investigation of the nonlinear effects, we apply a decomposition into a linear model component and the nonlinear deviation from the linear model for each effect based on the mixed model representation as discussed in Sect. 8.1.9. This has the advantage that the boosting algorithms compares linear and nonlinear effects of a continuous covariate in each iteration and only selects the nonlinear effect if it is really required. Note that the nonlinear effect should be defined as the deviation from the linear effect such that it does not comprise the linear effect as a special case. This allows for a proper discrimination, while using a naive penalized spline specification would always select the nonlinear effect since it contains additional flexibility as compared to the linear effect.
- For the spatial effect, we simultaneously included the dummy-coded regional variable, a Markov random field for the districts, and i.i.d. random effects for the districts as competitors within the same model. Since boosting will include function selection along with estimation, this allows us to determine automatically whether only one of the three types of spatial effects is required or whether a combination of two or even all three types of spatial effects is preferable. This is in contrast to the analyses in the previous section, where we exclusively considered one spatial modeling possibility at a time.

To further improve the selection properties, we chose the smoothing parameters in the penalized least squares base-learners such that each base-learner is assigned exactly one degree of freedom. This effectively makes each base-learner comparable to a linear effect of one single covariate. Note that we also have to use a ridge-penalized base-learner for categorical covariates with more than two categories. See [Hofner, Hothorn, Schmid, and Kneib \(2012\)](#) for details.

Figure 9.21 shows the cross-validated prediction error for this model specification for 10-fold cross validation. The minimum is reached within 7,563 iterations based on a total of 10,000 initial iterations. Note, however, that the risk is already very flat much earlier, and therefore the prediction performance seems to be rather insensitive to the exact values of the boosting iterations. In the following, we will present all results based on the model restricted to the optimal 7,563 iterations.

The main advantage of the boosting approach is its automated variable and function selection ability. Table 9.5 therefore shows the selection frequencies over the boosting iterations (second column) obtained for the different model terms. Any term not included in the table has never been selected during the first 7,563 iterations

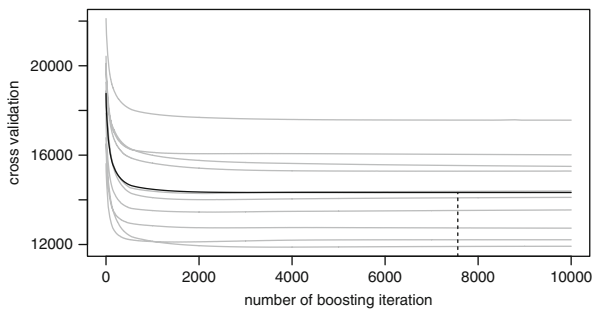


Fig. 9.21 Malnutrition in Zambia: cross validation criterion (*black line*) as a function of the boosting iterations. The *grey lines* indicate the prediction error obtained within a single cross validation fold. The *dashed vertical line* shows the optimal boosting iteration, i.e., the minimal cross validation criterion

Table 9.5 Selection frequencies in the boosting iterations and in 50-fold cross validation for the model comprising Markov random field and random effects

Variable / Term	Iterations	CV folds
$f(c_age, c_breastf)$	0.452	1.00
random effects	0.333	1.00
$m_education$	0.073	1.00
Markov random field	0.038	0.74
$region$	0.033	1.00
$f(m_bmi)$	0.028	1.00
$f(m_agebirth)$	0.014	0.06
c_gender	0.008	1.00
$f(m_height)$	0.007	0.54
$m_agebirth$	0.007	1.00
m_work	0.002	0.74
m_height	0.002	1.00
m_bmi	0.001	1.00

and is therefore dropped from the model. We can summarize the main findings from this table as follows:

- All three types of spatial effects are selected by the boosting algorithm. Hence, none of the three effects seems to be able to represent the spatial correlations present in the data to a satisfactory extent when being considered as the sole representative for the spatial effect.
- All three continuous covariates are included with both linear and nonlinear effects. This seems to indicate that nonlinear modeling is required in all three cases, which is in contrast to our findings from the previous paragraph, where at least mother’s age at birth and height of the mother were identified to be adequately represented by linear effects. However, the selection frequencies of both these covariates are rather small and the estimated effects are still very close to linear (not shown).

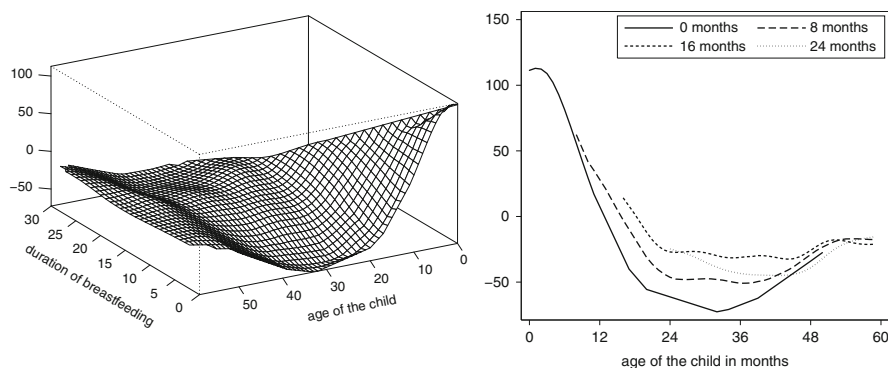


Fig. 9.22 Malnutrition in Zambia: interaction of the child's age and duration of breast-feeding obtained with boosting (*left*). The *right panel* shows the age effect for various durations of breast-feeding

- The interaction between age of the child and duration of breast-feeding (visualized in Fig. 9.22) is clearly identified with rather high selection frequency. The estimated function is much smoother than in the previous section, but still recovers the same general structure.
- Both categorical covariates are included in the model.

We close the discussion of Table 9.5 with a word of caution: though it is tempting to interpret the selection frequencies as a measure of importance for the corresponding term, this is not formally justifiable. The reason is that early inclusions typically contribute more strongly to the reduction of the risk, while later inclusions often only marginally improve the risk. Correspondingly, a variable may be selected very often in late boosting iterations and may still contribute only a negligible part to the overall estimated predictor. On the other hand, an important covariate may be selected in a relatively small number of early iterations and may still be very important for the complete model fit. As a consequence, a combination of the selection frequencies and the estimated effects should be interpreted to get an impression of the importance of the various estimated effects. As a further indication of a variable's importance, we computed 50-fold cross validation results with a small number of only 2,000 boosting iterations and included the frequencies for inclusion of an effect in the 50 folds in column three of Table 9.5. While we do not try to optimize the number of boosting iterations but rather use a small number of iterations, we will be able to identify those variables that are selected early in the boosting algorithm and are therefore important for predicting the nutritional status. In fact, most of the effects selected by the boosting iteration are also selected in all of the 50 folds but some variables seem to be of somewhat minor importance. In particular, the nonlinear effects of age at birth and height of the mother are ranked lower, which is in accordance with the selection results from the previous section. In addition, the Markov random field has smaller selection frequency than the random

Table 9.6 Malnutrition in Zambia: selection frequencies in the boosting iterations and in 50-fold cross validation for the model comprising only i.i.d. random effects for the spatial effect

Variable / Term	Iterations	CV folds
Random effects	0.504	1.00
$f(c_age, c_breastf)$	0.270	1.00
$m_education$	0.123	1.00
$region$	0.068	1.00
c_gender	0.015	1.00
$m_agebirth$	0.007	1.00
m_height	0.007	1.00
m_bmi	0.003	1.00
m_work	0.003	0.74
$f(m_bmi)$	0.001	1.00
$f(m_height)$	–	0.54
$f(m_agebirth)$	–	0.06



Fig. 9.23 Malnutrition in Zambia: estimated spatial effects obtained with boosting for the three different model specifications comprising Markov random field and random effect (*top left*), Markov random field only (*top right*), and random effects only (*bottom*)

effects term which may give some indication that an unstructured spatial effect is more important than a structured effect.

As a variation of our initial model we re-estimated two more models with different specifications for the spatial effect. More precisely, we include only the Markov and random field or the i.i.d. district-specific random effects for the district-level spatial information and not the combination of the two. For the two resulting models, the optimal boosting iterations were determined as 21,823 (Markov random fields) and 2,723 (i.i.d. random effects) via 10-fold cross validation. Especially in the latter case, a much simpler model specification arises as shown by the selection frequencies in Table 9.6. In this case, the chosen model no longer comprises nonlinear effects of mother's height and age at birth, while a slightly nonlinear effect remains for the body mass index of the mother. This model is therefore even closer to the one identified in the previous section. When considering selection frequencies from a 50-fold cross validation again, we find a similar picture. While most effects identified by the boosting algorithm are included in all 50 folds, the nonlinear effects of age at birth and height of the mother are only selected in only a relatively small number of folds.

Figure 9.23 shows the estimated spatial effects on district level for the three different model variants. In case of the first model comprising the Markov random field and i.i.d. random effects, the sum of both effects is shown to facilitate the comparison between the models. Obviously, all models identify the same spatial structure especially when comparing the combined model with the model consisting of only the Markov random field. The model containing only i.i.d. random effects also shows a similar spatial distribution of the estimated effects, but the magnitude of the effects is somewhat lower (which most probably will also be related to the much smaller number of boosting iterations identified for this model).

9.9 Bibliographic Notes

The literature references about nonparametric regression provided in Sect. 8.4 of Chap. 8 remain valid for this chapter. We will only briefly summarize the key references which specifically treat additive models and their extensions.

The classical textbook about GAMs (which is still worth reading) is that of [Hastie and Tibshirani \(1990\)](#). [Marx and Eilers \(1998\)](#) developed P-spline strategies to directly fit GAMs (without backfitting), while [Eilers and Marx \(2002\)](#) presented P-spline variants of STAR models, referred to as generalized linear additive smooth structures (GLASS). [Wood \(2006\)](#) describes GAMs based on P-splines and their application using the R package `mgcv`. [Ruppert et al. \(2003\)](#) describe GAMs and their extensions in a unified approach as mixed models. Estimation is based on methods and software for this model framework. [Fahrmeir and Kneib \(2011\)](#) provide a recent overview on Bayesian smoothing approaches for longitudinal, spatial, and event history data comprising both fully Bayes and empirical Bayes methods.

[Fan and Gijbels \(1996\)](#), [Loader \(1999\)](#), and [Härdle et al. \(2004\)](#) propose approaches based on local smoothers. The monograph by [Fotheringham, Brunsdon,](#)

and Charlton (2002) describes models with spatially varying coefficients where estimation is also based on local smoothers. The book by Yatchew (2003) describes the use of additive models in econometrics. The mathematically demanding monograph by Gu (2002) imbeds additive models into the theory of reproducing Hilbert spaces. Finally, Denison, Holmes, Mallick, and Smith (2002) describe Bayesian adaptive regression splines as building blocks for GAMs and their extensions. The book by Hastie et al. (2009) on “The Elements of Statistical Learning” provides a lucid presentation of contemporary statistical techniques, including boosting.